

ELL 100 – Introduction to Electrical Engineering

LECTURE 5: CIRCUIT ANALYSIS

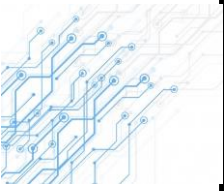
KIRCHOFF'S LAWS AND MESH ANALYSIS

Outline

- ☐ Introduction
- ☐ Voltage Division in Series Circuit
- ☐ Current Division in Parallel Circuit
- ☐ Kirchhoff's Laws
- ☐ Mesh Analysis
- ☐ Super Mesh Concept
- ☐ Exercise/Numerical Analysis



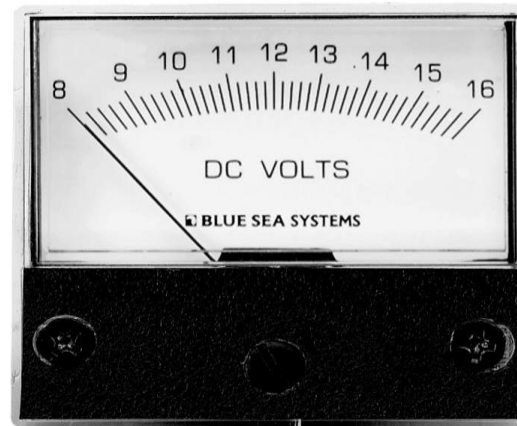
INTRODUCTION



- Current gets divided in different path when resistances are connected in parallel, hence **current division** is applicable in **parallel circuit**.
- Voltage gets divided in series but current remain same, hence **voltage division** is applicable in **series circuit**.
- Currents and voltages follow **KCL** and **KVL** laws in circuits at junction points, nodes and in loops.
- **Complex electrical circuits** with **multiple loops** can be solved via **mesh analysis**.

INTRODUCTION

- Current/Voltage Division finds applications in construction of **electric meters and sensor devices**.



INTRODUCTION

- **Voltage division** is used in reducing voltage level to lower value using Voltage Divider.

NANR®

A 200kV voltage Divider with ratio 10000:1



INTRODUCTION

- A 600 kV high voltage capacitive voltage divider.



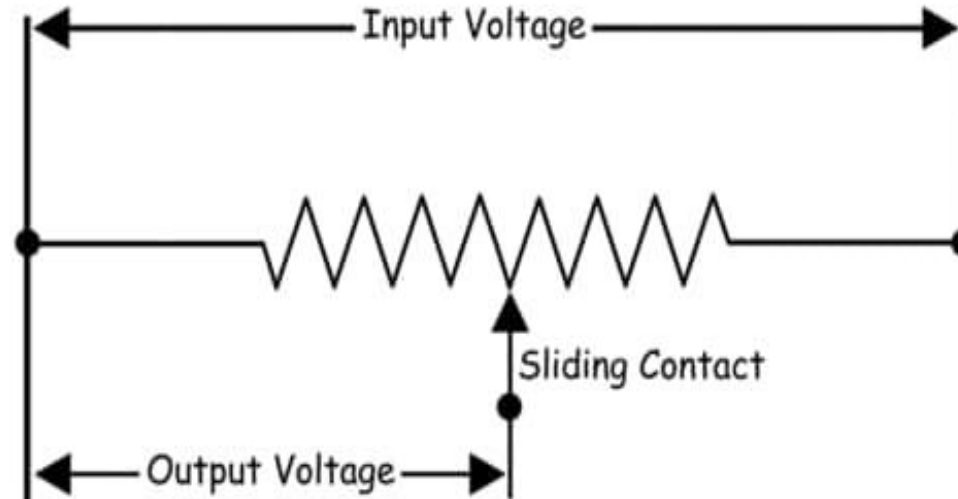
INTRODUCTION

- **Resistive voltage dividers are used for measurements of lightning impulses and other large impulses voltages.**



INTRODUCTION

- **Voltage division** concept is used in making rheostat, potentiometers



INTRODUCTION

- **Voltage division** concept is used in making radios, amplifiers and electronic devices for adjusting signal levels.



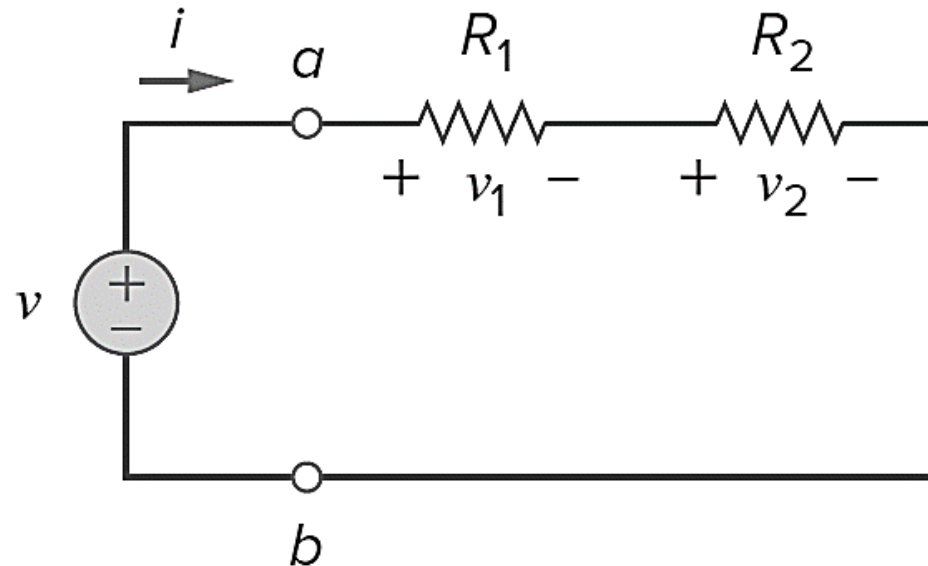
INTRODUCTION

- Principal of voltage division and current division is used in potential transformers and current transformers for stepping down high voltage and high current in power plants.



VOLTAGE DIVISION

- The total **voltage** gets **divided** in **series circuit** and while the **current** flowing through every element is the **same**.
- The voltage division rule can be understood by considering a series circuit shown below.



VOLTAGE DIVISION

- The voltages across series resistors are,

$$V_1 = IR_1 \quad (4)$$

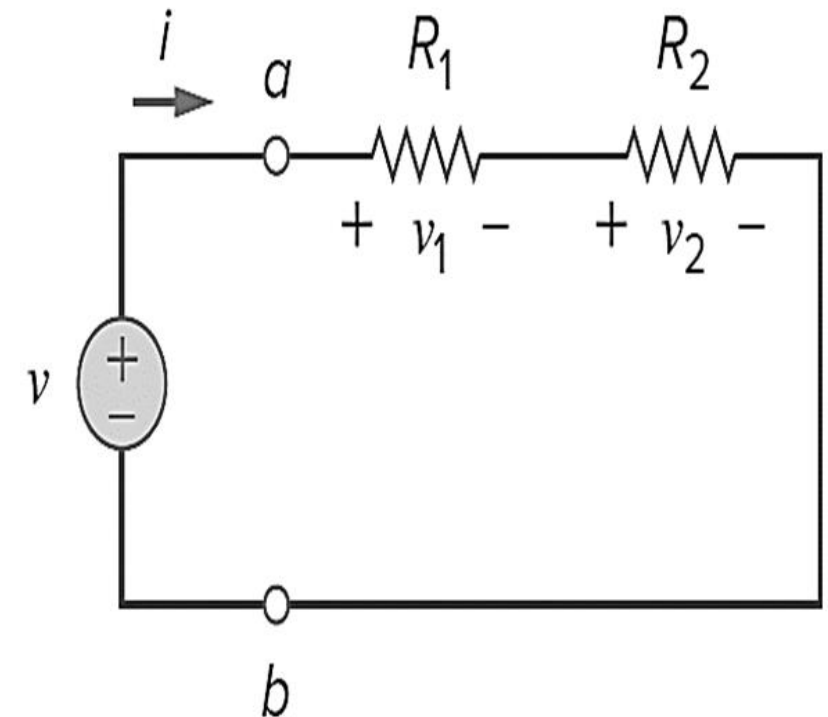
$$V_2 = IR_2 \quad (5)$$

- Putting (3) in (4), Voltage drop across R_1

$$V_1 = \frac{R_1}{R_1 + R_2} V \quad (6)$$

- Putting (3) in (5), Voltage drop across R_2

$$V_2 = \frac{R_2}{R_1 + R_2} V \quad (7)$$





VOLTAGE DIVISION

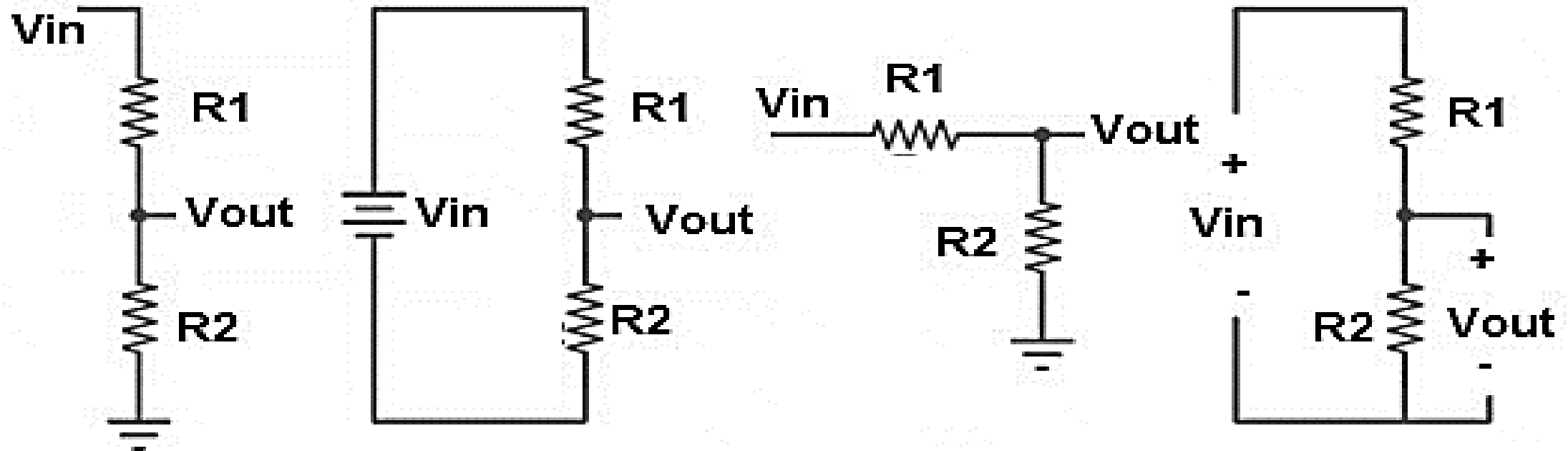


- V is divided among the resistors in direct proportion to their resistance. Larger the resistance, larger is voltage drop. This is called the principle of **voltage division**.
- For N resistors in series, voltage across nth resistor is given by,

$$V_n = \frac{R_n}{R_1 + R_2 + R_3 + \dots R_n} V$$

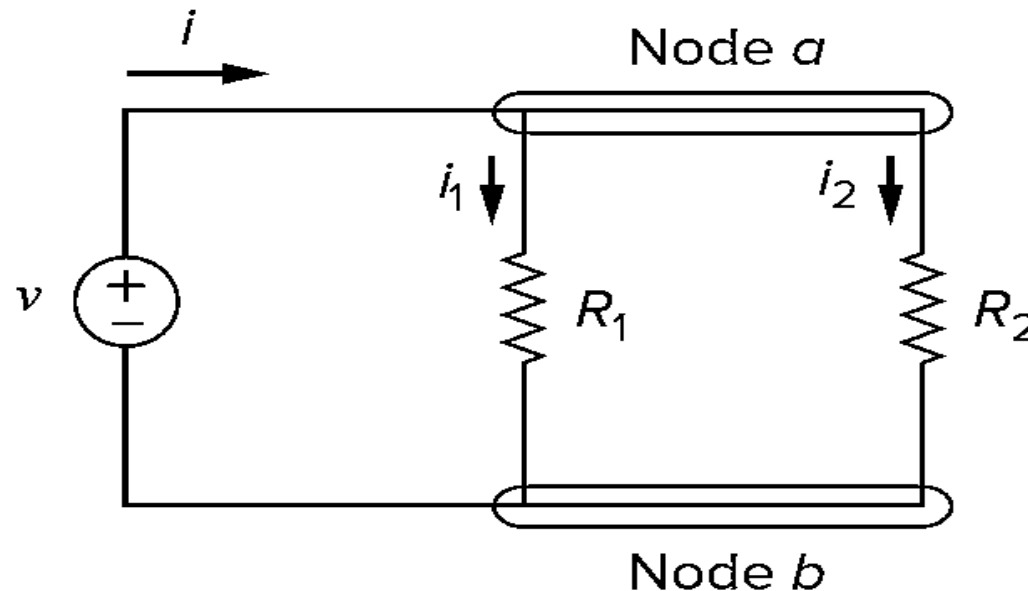
VOLTAGE DIVISION

- Different types of voltage division circuits are,



CURRENT DIVISION

- The total **current** gets **divided** in **parallel circuit** while the **voltage** across each element is the **same**.
- The current division rule can be understood by considering a parallel circuit shown below.





CURRENT DIVISION



$$i_1 = \frac{IR_2}{R_1 + R_2}$$

$$i_2 = \frac{IR_1}{R_1 + R_2}$$

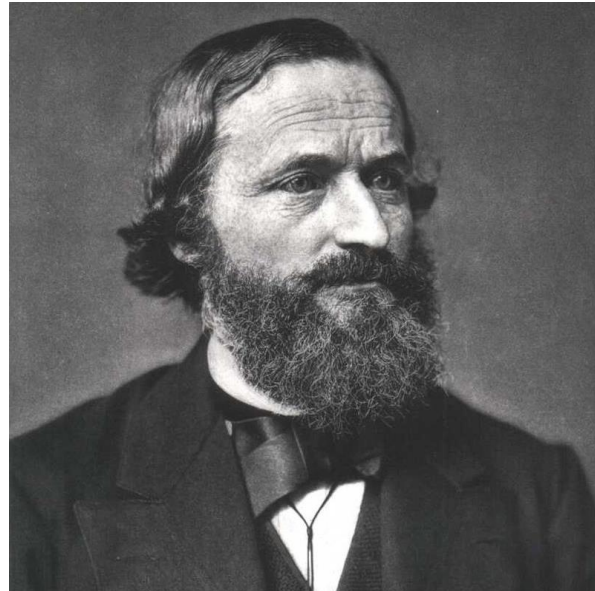
- Total current I is shared by the resistors in inverse proportion to their resistances. This is known as the principle of current division.
- In terms of conductances we have,

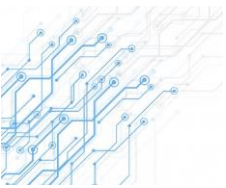
$$i_1 = \frac{IG_1}{G_1 + G_2}$$

$$i_2 = \frac{IG_2}{G_1 + G_2}$$

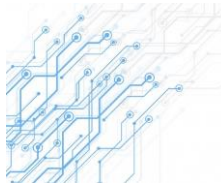
KIRCHHOFF'S LAWS

- Kirchhoff's laws are two equalities that deal with the current and voltage in electrical circuits.
- They were first described in 1845 by German physicist Gustav Kirchhoff.





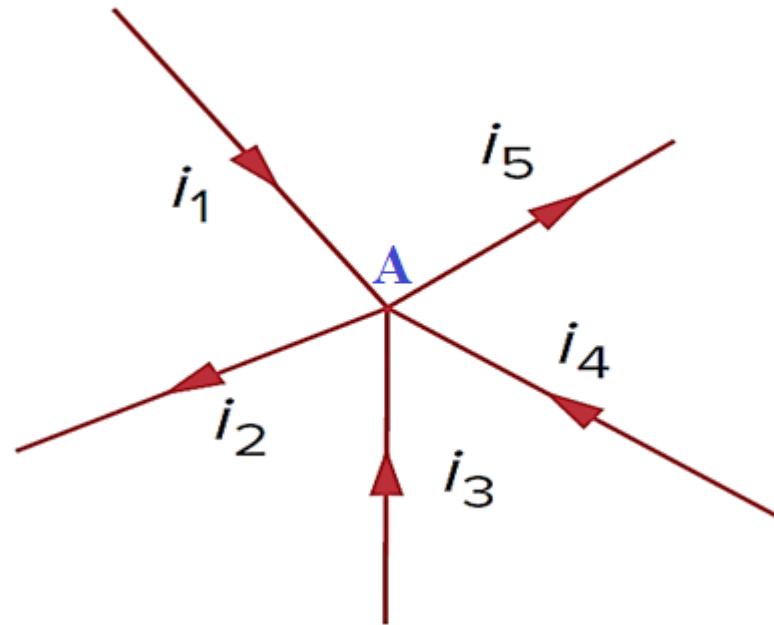
KIRCHHOFF'S LAWS



- Ohm's law by itself is not sufficient to analyze circuits.
- When it is coupled with **Kirchhoff's laws**, we have a sufficient, powerful set of tools for analyzing a large variety of electric circuits.
- Kirchhoff's current law is based on the law of conservation of **charge**.
- Kirchhoff's voltage law is based on the law of conservation of **energy**.
- Kirchhoff's laws can be applied to **AC** circuits in the same way as for **DC** circuits provided we use **phasor** sums rather than **algebraic** sums.

KIRCHHOFF'S CURRENT LAW (KCL)

- Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.
- Fig. below shows that currents are entering and leaving the **Node A**.

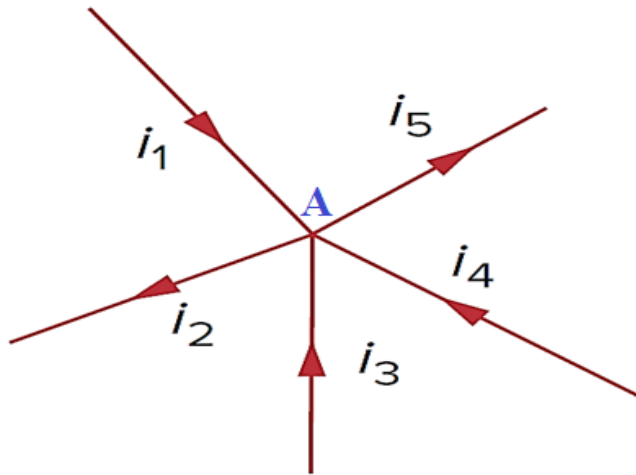


KIRCHHOFF'S CURRENT LAW (KCL)

- Before applying **KCL** sign convention is very important. The following sign convention is used:

Current Entering node "-ve"
Current Leaving node "+ve"

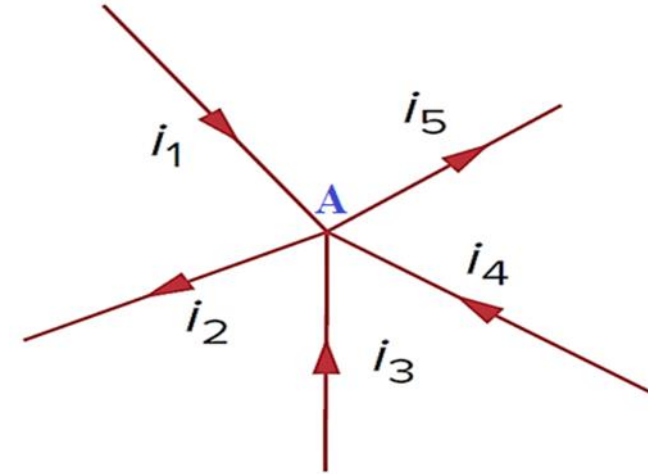
- Applying **Kirchhoff's current law** with sign convention one has,



$$-I_1 + I_2 - I_3 - I_4 + I_5 = 0$$

KIRCHHOFF'S CURRENT LAW (KCL)

$$I_1 + I_3 + I_4 = I_2 + I_5$$

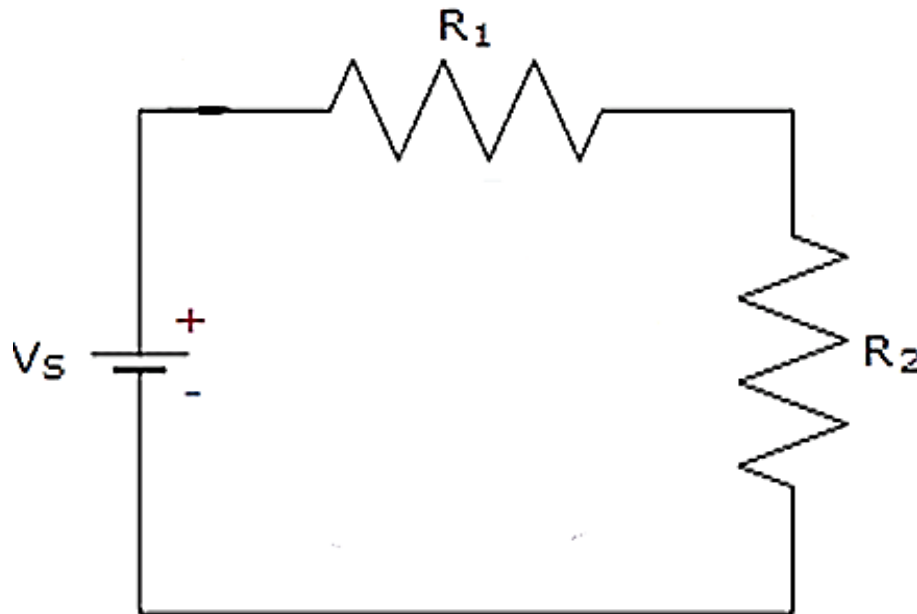


➤ Alternatively **Kirchhoff's current law** is also stated as,

The sum of the currents entering a node is equal to the sum of the currents leaving the node.

KIRCHHOFF'S VOLTAGE LAW (KVL)

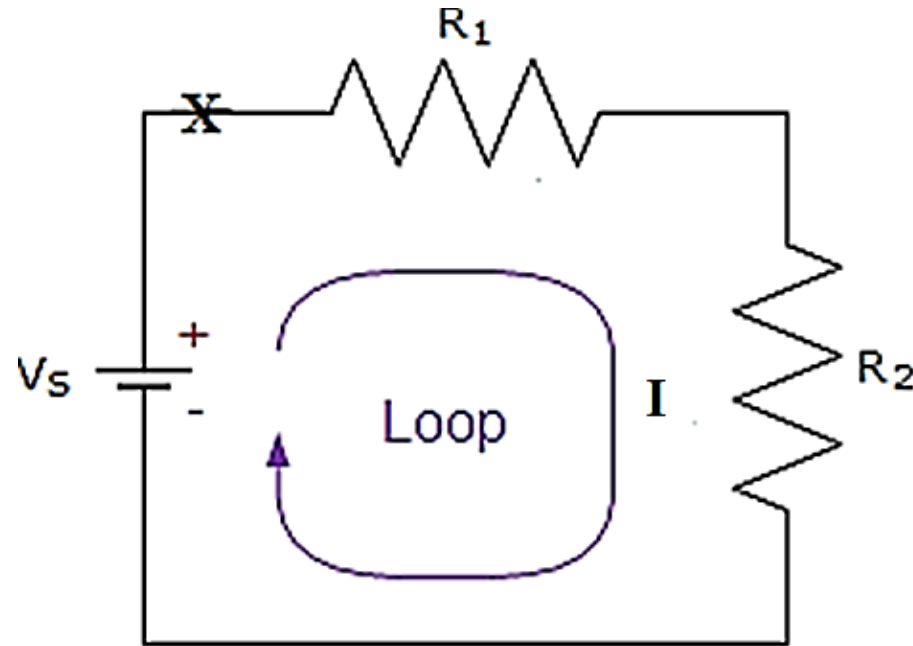
- Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.
- To illustrate KVL, consider the circuit.



KVL

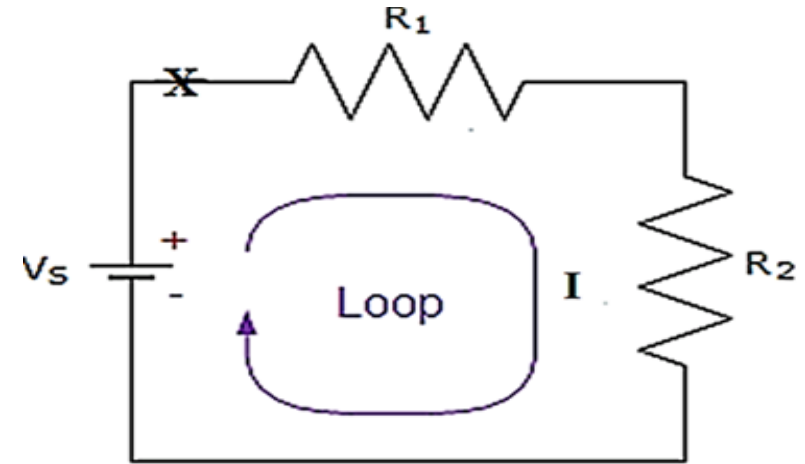
Step 1: Let us assume clockwise current in the loop.

Step 2: Moving in direction of current and writing KVL equation.
Let us assume we start moving from location **X** towards right.



KVL

$$V_S = IR_1 + IR_2$$



If voltage drop across resistances are termed as V_1 and V_2 ,

$$V_S = V_1 + V_2$$

Furthermore,

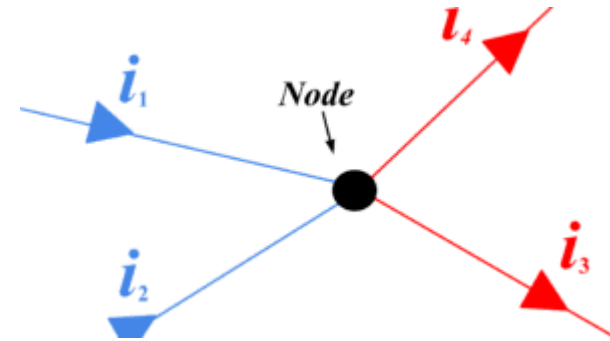
$$V_S - V_1 - V_2 = 0$$

shows total voltage is sum of drops across each resistances which is known as **Kirchhoff's Voltage Law**.

KCL AND KVL

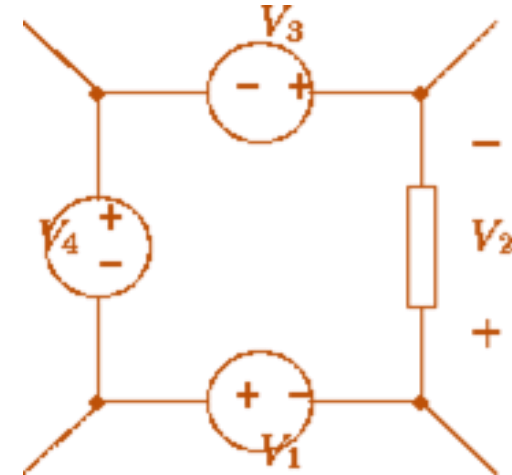
➤ Mathematically KCL is written as,

$$\sum_{n=1}^N I_n = 0$$



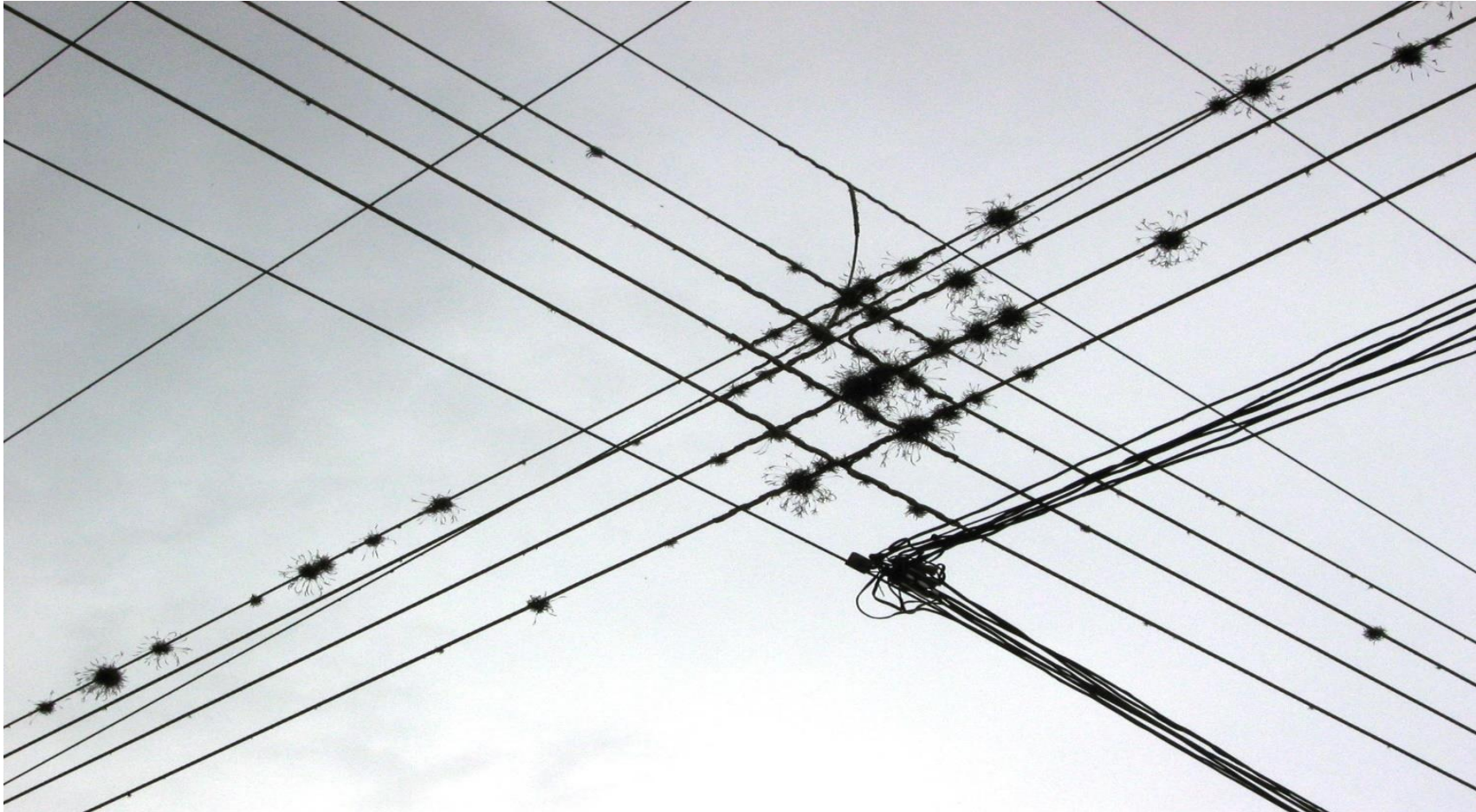
where, N is number of branches, I_n is nth branch current entering/leaving the node. Similarly KVL for voltage across total M elements,

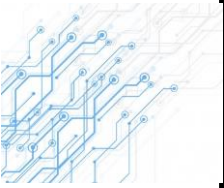
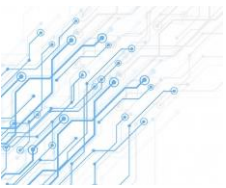
$$\sum_{m=1}^M V_n = 0$$



MESH ANALYSIS

- **Mesh analysis** helps us to solve complex electrical networks.



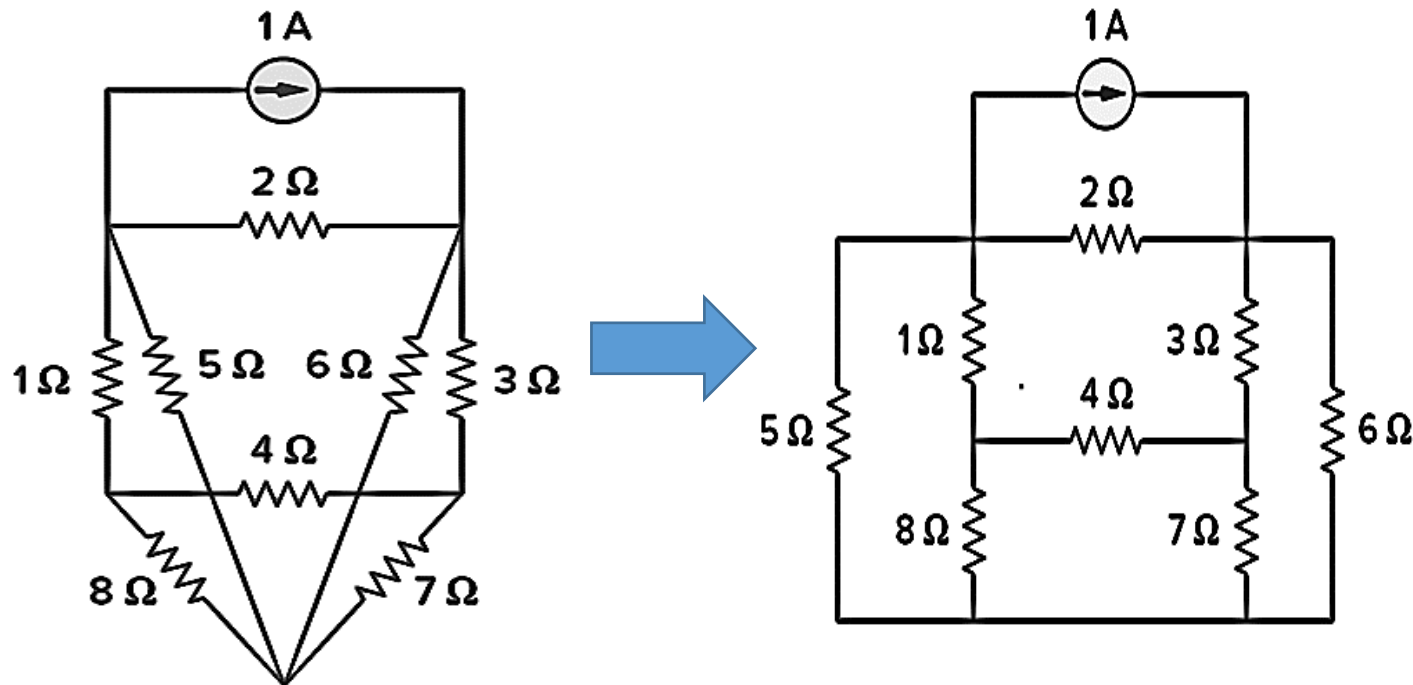


MESH ANALYSIS

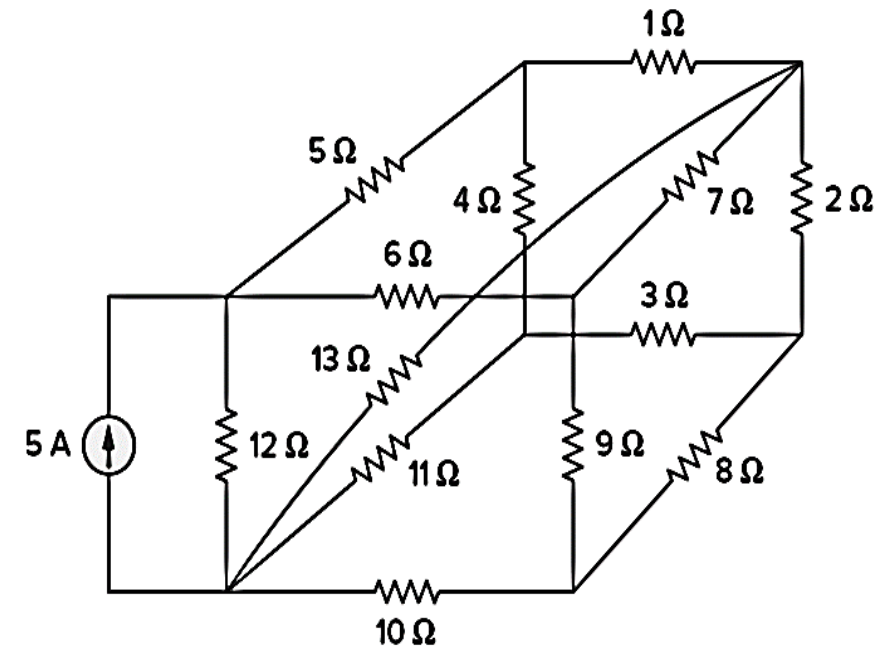
- **Loop:** It is a closed path with no node passed more than once.
- **Mesh:** A mesh is a loop that does not contain any other loop within it.
- **Mesh analysis** is only applicable to a **circuit** that is **planar**.
- **Planar Circuit:** A planar circuit is one that can be drawn in a plane with no branches crossing one another.
- **Non-Planar Circuit:** A nonplanar circuit is one that can be drawn in a plane with branches crossing one another.

MESH ANALYSIS

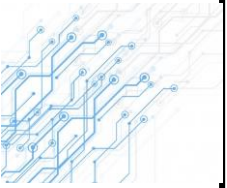
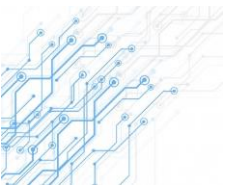
- Some circuits may have crossing branches, but they can be redrawn as planar.



PLANAR CIRCUIT



NONPLANAR CIRCUIT

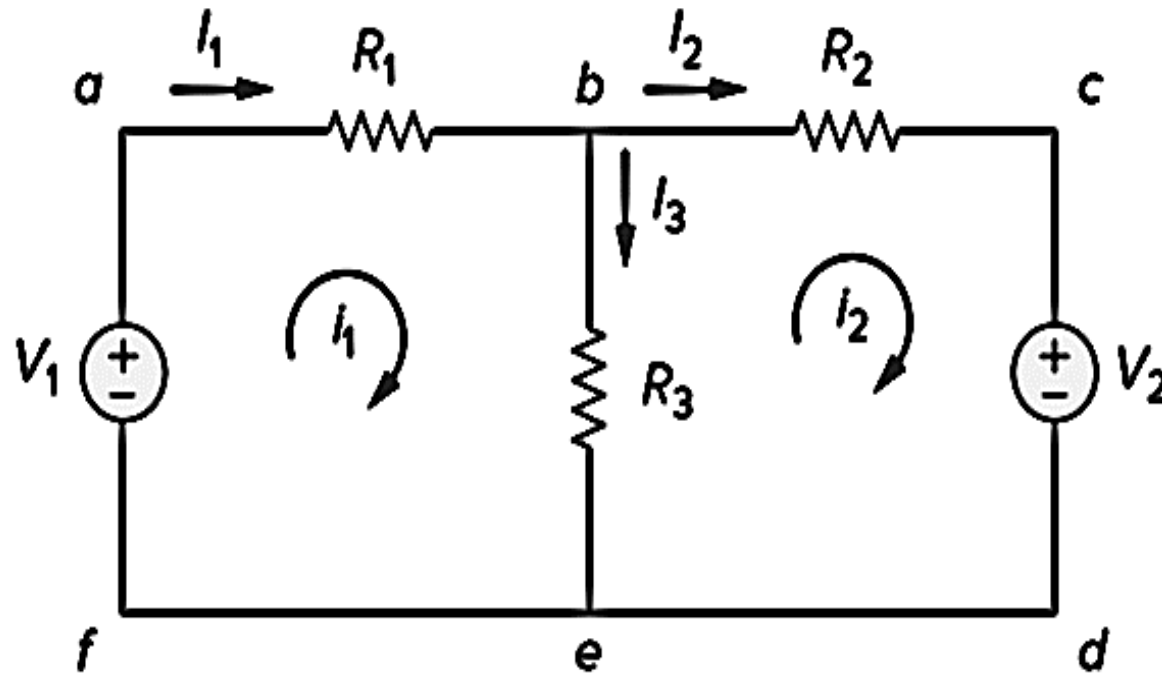


MESH ANALYSIS

- Let us learn mesh analysis to planar circuits that do not contain current sources.
- In the mesh analysis of a circuit with n meshes, we take the following three steps:
 - Assign mesh currents $i_1, i_2, \dots$, in to the n meshes.
 - Apply **KVL** to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
 - Solve the resulting n simultaneous equations to get the mesh currents.

MESH ANALYSIS

- Consider circuit shown below with two meshes, currents are assigned in clockwise direction in mesh. Let us analyze mesh for calculating current using KVL in each mesh.



MESH ANALYSIS

- Applying KVL in mesh 1.

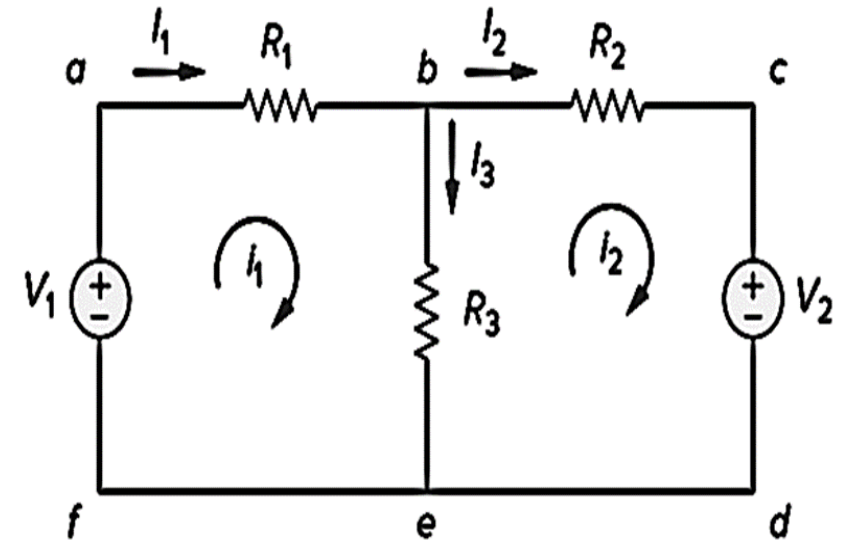
$$-V_1 + R_1 i_1 + R_3 i_3 = 0 \quad (1)$$

- Applying KCL at node b,

$$-i_1 + i_2 + i_3 = 0 \Rightarrow i_3 = i_1 - i_2 \quad (2)$$

- Substituting (2) in (1) ,

$$V_1 = (R_1 + R_3) i_1 - R_3 i_2 \quad (I)$$



MESH ANALYSIS

- Applying KVL in mesh 2.

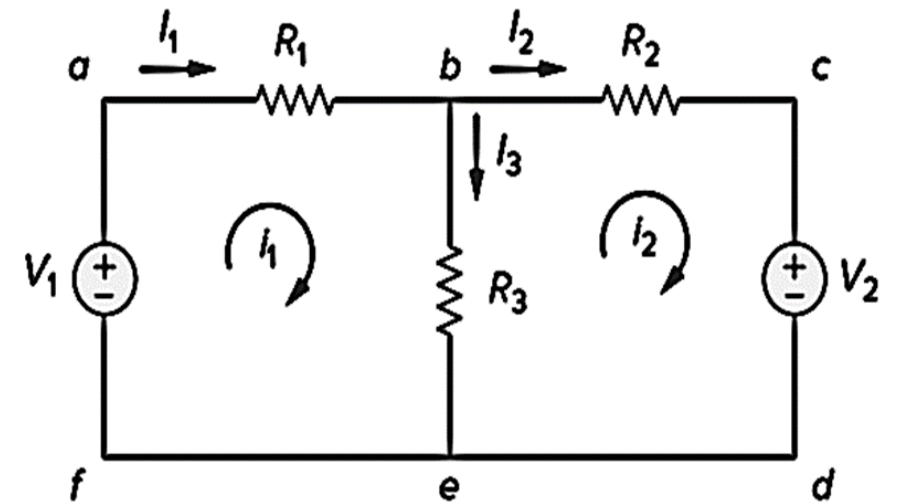
$$R_2 i_2 + V_2 - R_3 i_3 = 0 \quad (1)$$

- Replacing i_3 ,

$$R_2 i_2 + V_2 - R_3 (i_1 - i_2) = 0 \quad (2)$$

- Substituting (2) in (1) ,

$$V_2 = -R_3 i_1 + (R_1 + R_2) i_2 \quad (II)$$



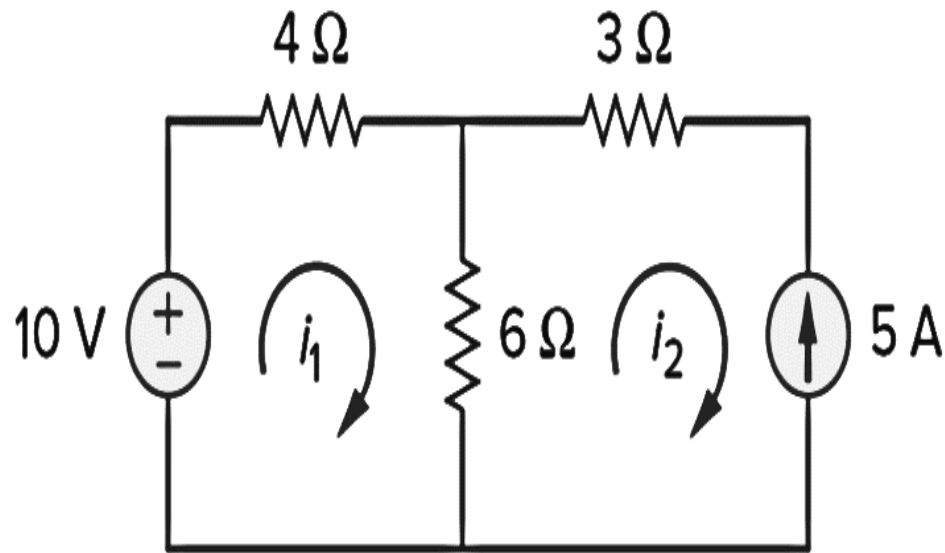


MESH ANALYSIS

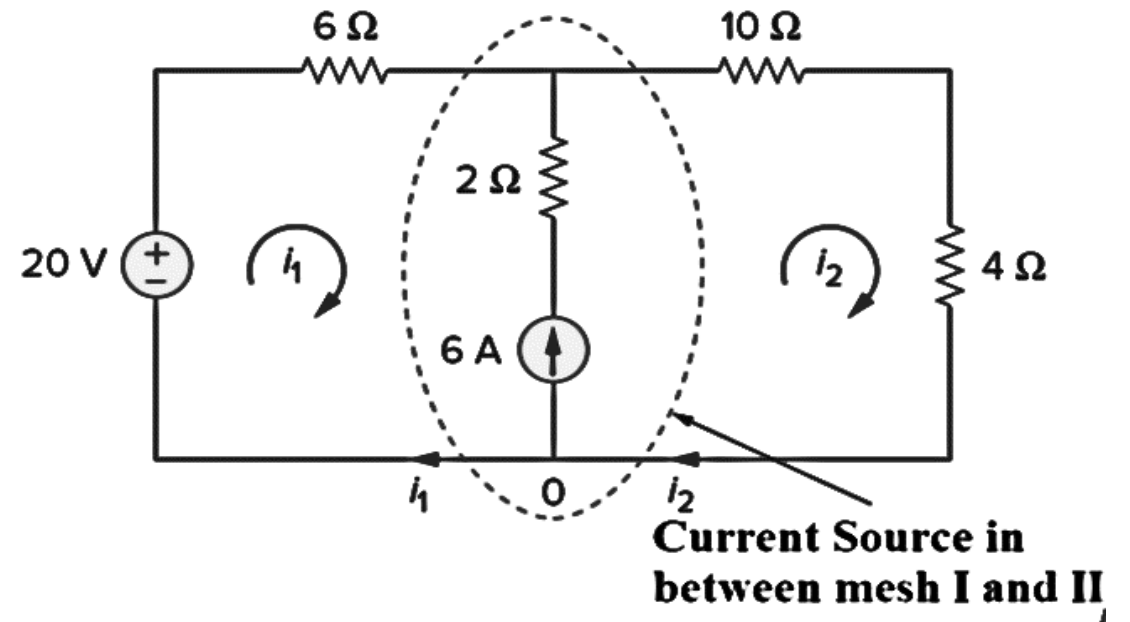
- Eqs. (I) and (II) are mesh 1 and 2 equations.
- These can be solved to obtain two unknowns i_1 and i_2 .
- Moreover, i_3 can also be obtained which is difference of two.
- Hence, two mesh circuit currents are calculated when voltage sources and resistance values are given using steps of mesh analysis.
- In case of three meshes in a circuit there are three equations and we can use Cramer's rule to solve 3 linear equations.

MESH ANALYSIS

- Let us learn mesh analysis to planar circuits that contain **current sources**.
- Two cases arises in these circuits,



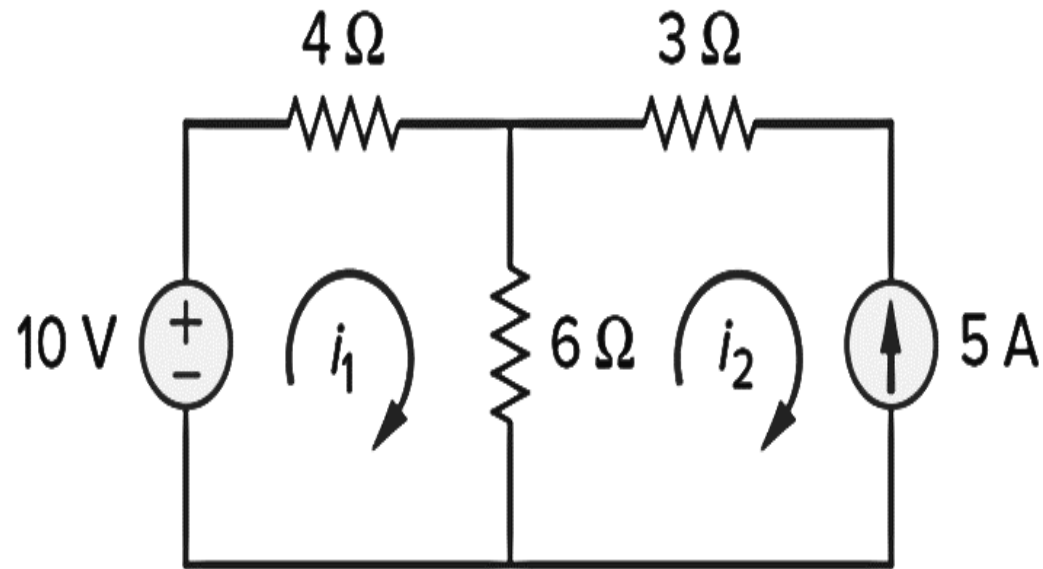
Case 1: Current source exist in arm of a mesh (here in II mesh).



Case 2: Current source exist in between meshes.

MESH ANALYSIS

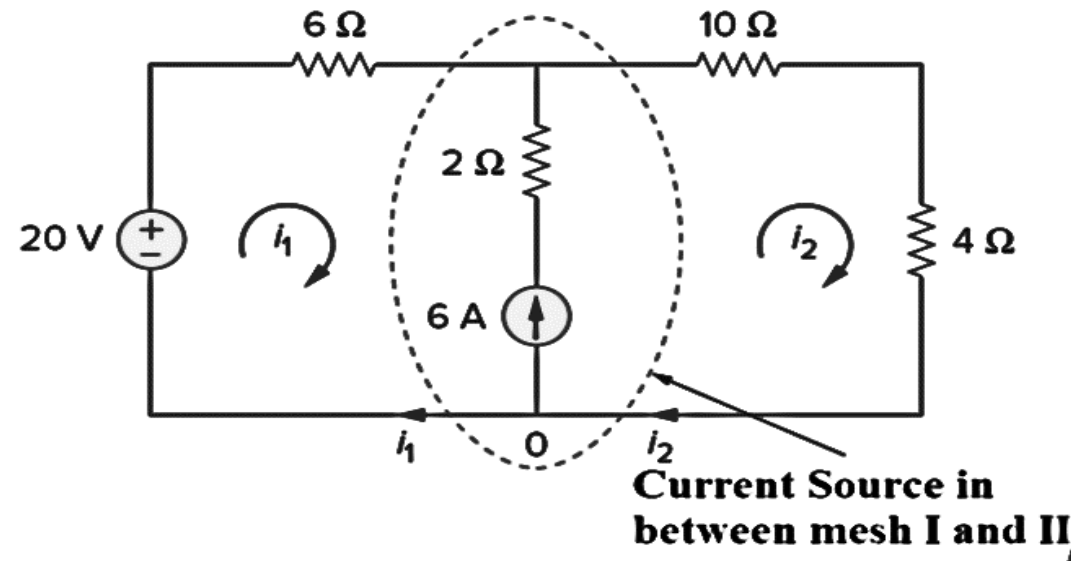
- **Case 1** makes calculation easier as current source itself give value of current flowing in that mesh.



- Hence, i_2 becomes -5 A, putting value of i_2 in KVL, current of mesh I is easily calculated.

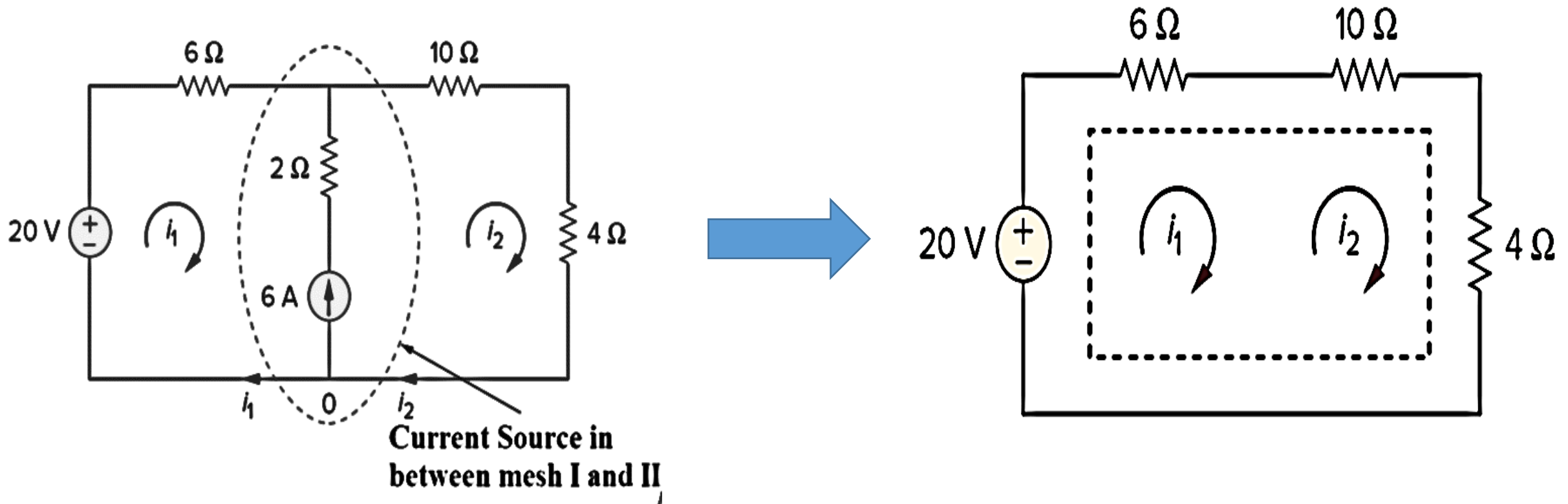
SUPER MESH CONCEPT

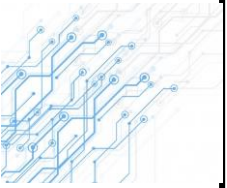
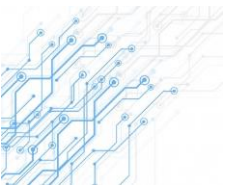
- When current source (dependent/independent) occurs between two meshes we apply super mesh concept to solve the circuit.
- We create a **super mesh** by **excluding the current source and any elements connected in series with it** while taking KVL.
- Let us solve such circuit shown below,



SUPER MESH CONCEPT

- Converting Fig. having two meshes into a circuit having single super mesh.





SUPER MESH CONCEPT

- Applying KVL to the supermesh.

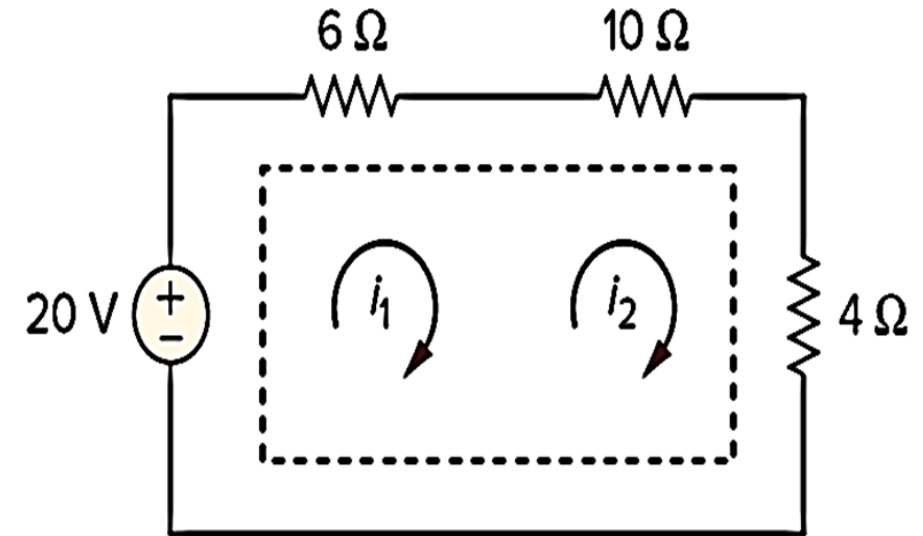
$$-20 + 6i_1 + 10i_2 + 4i_2 = 0$$

$$\Rightarrow \boxed{6i_1 + 14i_2 = 20}$$

- Applying KCL at bottom node,

$$\boxed{i_2 = i_1 + 6}$$

- We can easily solve above two equations to obtain i_1 and i_2 .
- A supermesh requires the application of both KVL and KCL.

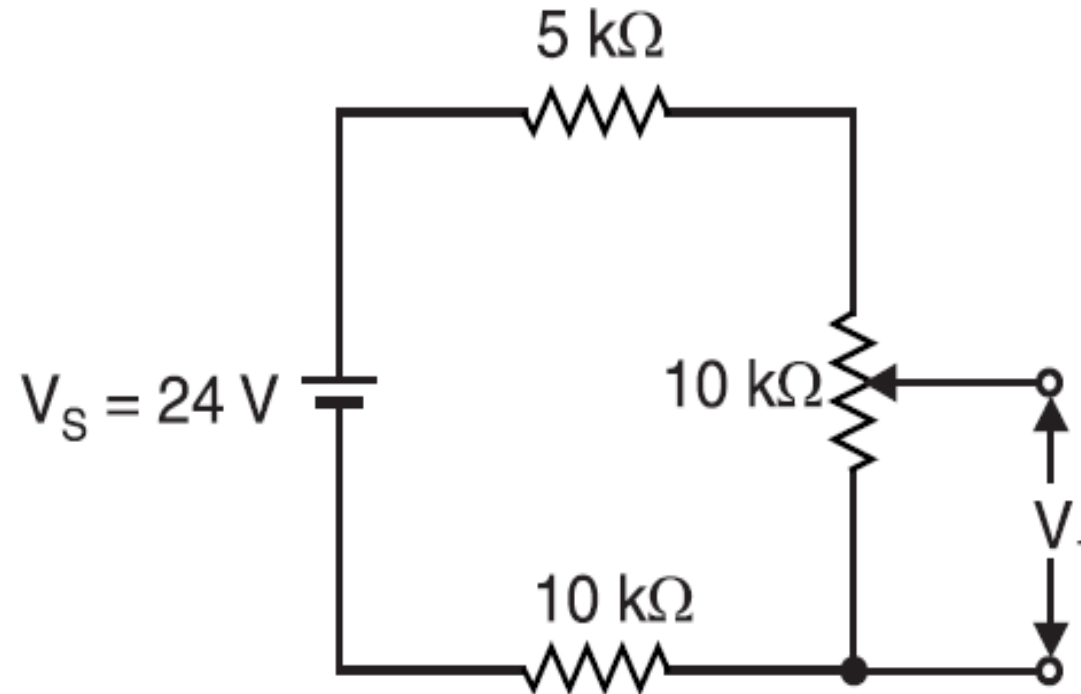




EXERCISE/NUMERICAL ANALYSIS



Q. A $10\text{ k}\Omega$ potentiometer connected in a series circuit as an adjustable voltage divider. What total range of voltage V_1 can be obtained by adjusting the potentiometer through its entire range.





EXERCISE/NUMERICAL ANALYSIS



Soln: Total circuit resistance is

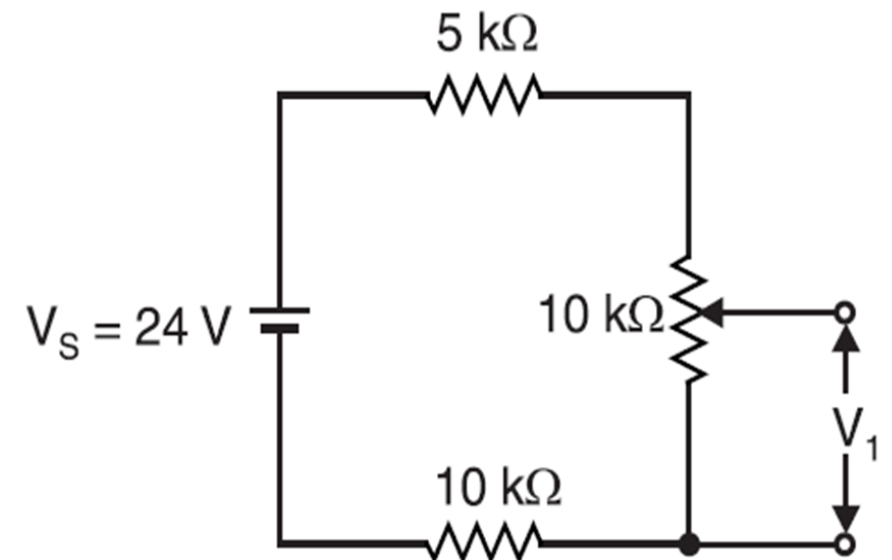
$$R_T = R_1 + R_2 + R_3 = 5 \text{ k}\Omega + 10 \text{ k}\Omega + 15 \text{ k}\Omega = 25 \text{ k}\Omega$$

The total voltage E that appears across the end terminals of potentiometer is,

$$V_P = \frac{R_2}{R_T} V_s = \frac{10}{25} \times 24 = 9.6 \text{ V}$$

When the wiper arm is at the top of the

potentiometer, $V_o = \frac{10}{10} \times 9.6 = 9.6 \text{ V}$





EXERCISE/NUMERICAL ANALYSIS



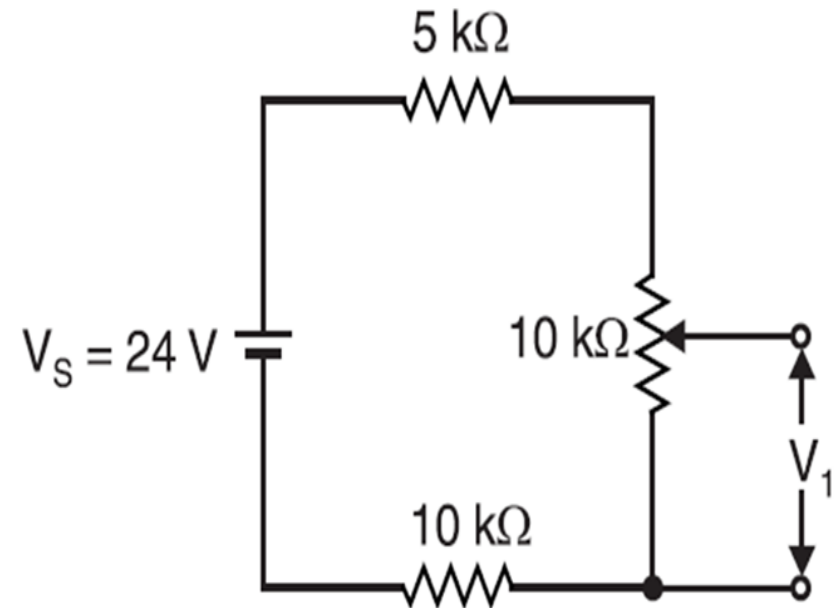
When the wiper arm is at bottom of the potentiometer

$$V_o = \frac{0}{10} \times 9.6 = 0 \text{ V}$$

When the wiper arm is in middle,

$$V_o = \frac{5}{10} \times 9.6 = 4.8 \text{ V}$$

Hence voltage divider helps us to obtain different voltages from fixed input of 24 V.

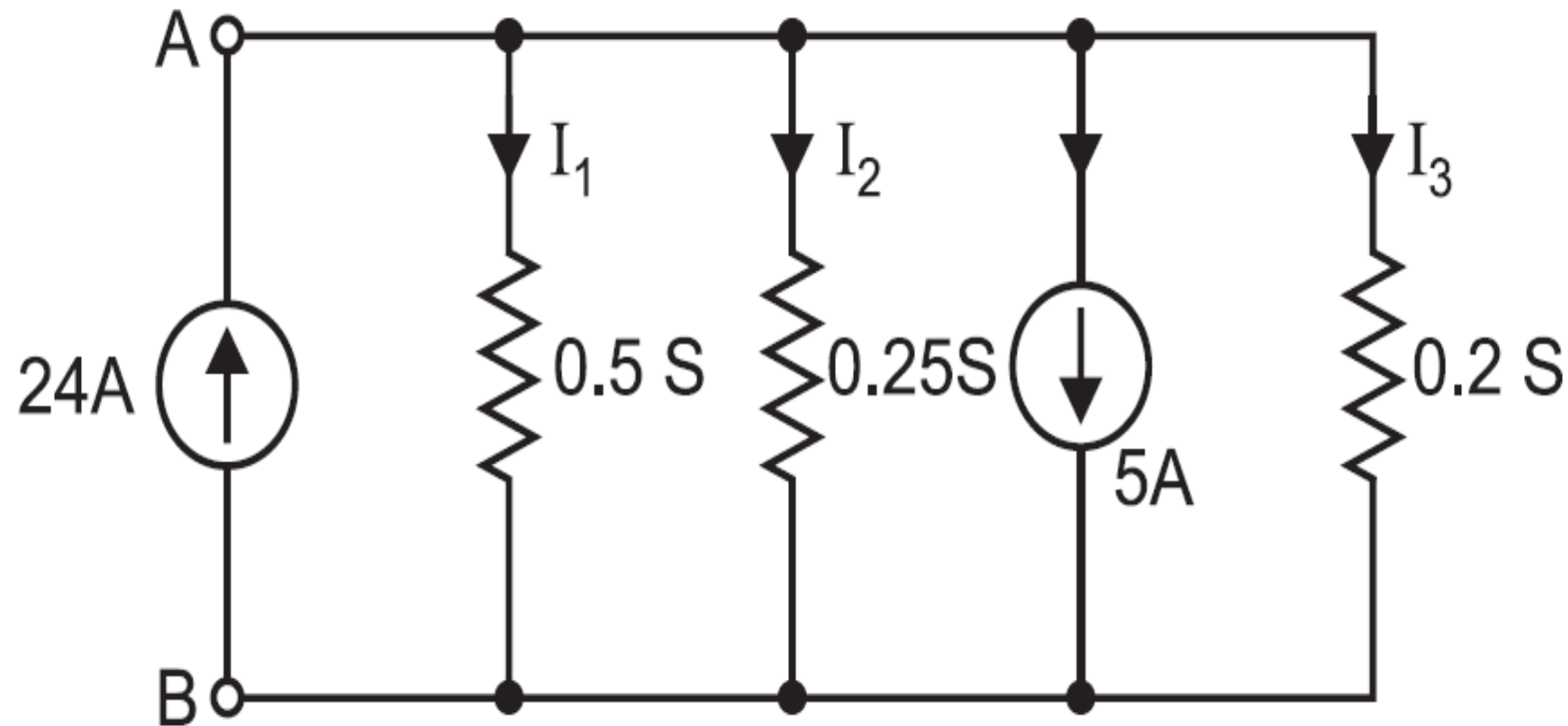




EXERCISE/NUMERICAL ANALYSIS



Q. Find the three branch currents in the circuit shown below. What is the potential difference between points A and B?





EXERCISE/NUMERICAL ANALYSIS



Soln: Current sources in parallel add algebraically.

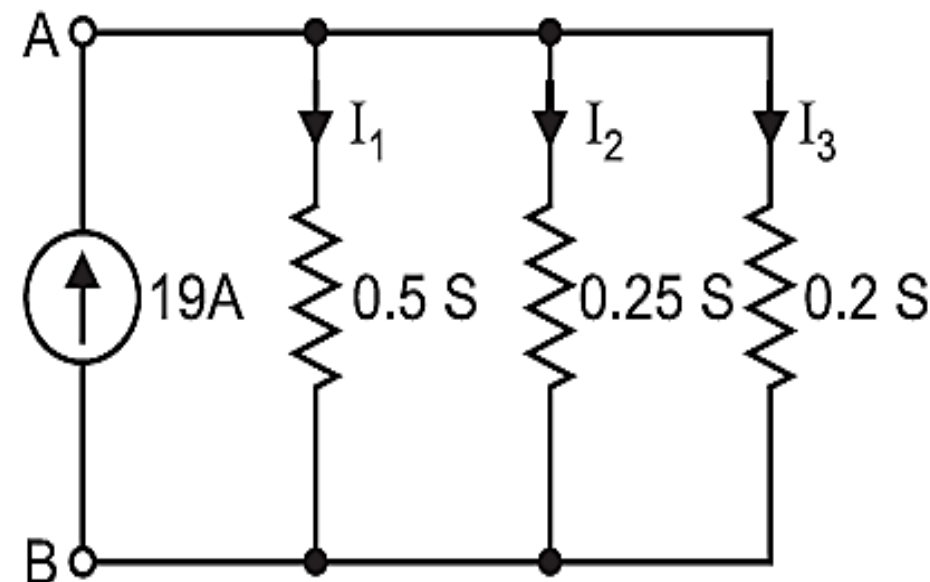
Therefore, the two current sources can be combined to give the resultant current source of current, $I = 24 - 5 = 19 \text{ A}$

Total circuit conductance is

$$\begin{aligned} G_T &= G_1 + G_2 + G_3 \\ &= .5 + 0.25 + 0.2 = 0.95 \text{ S} \end{aligned}$$

Applying Current Division Rule,

$$I_1 = \frac{G_1}{G_p} I = \frac{0.5}{0.95} \times 19 = 10 \text{ A}$$





EXERCISE/NUMERICAL ANALYSIS



$$I_2 = \frac{G_2}{G_p} I = \frac{0.25}{0.95} \times 19 = 5 \text{ A}$$

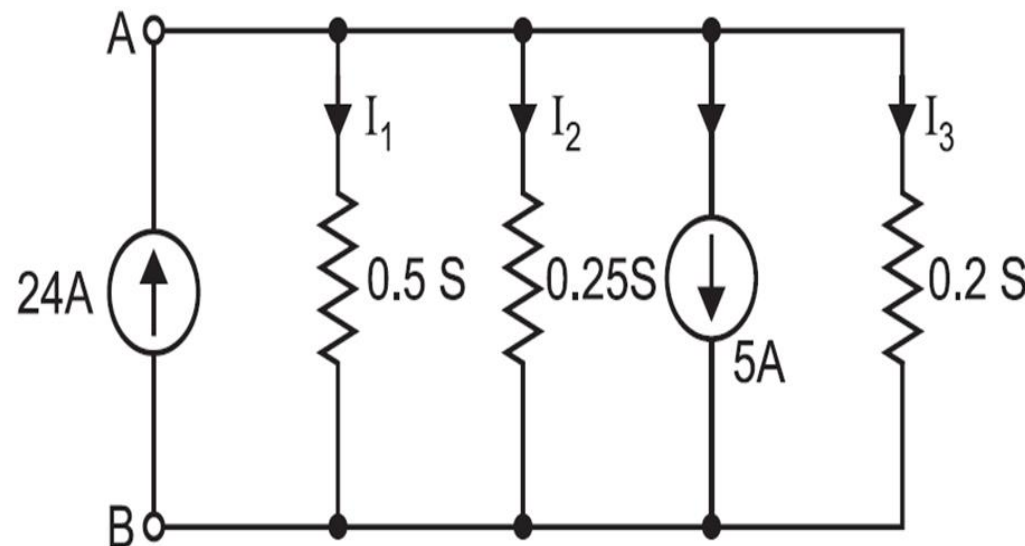
$$I_3 = \frac{G_3}{G_p} I = \frac{0.2}{0.95} \times 19 = 4 \text{ A}$$

As all branches are in parallel, voltage across each branch is same.

Hence,

$$V_{AB} = \frac{I_1}{G_1} = \frac{I_2}{G_2} = \frac{I_3}{G_3}$$

$$V_{AB} = \frac{10}{0.5} = \frac{5}{0.25} = \frac{4}{0.2} = 20 \text{ V}$$

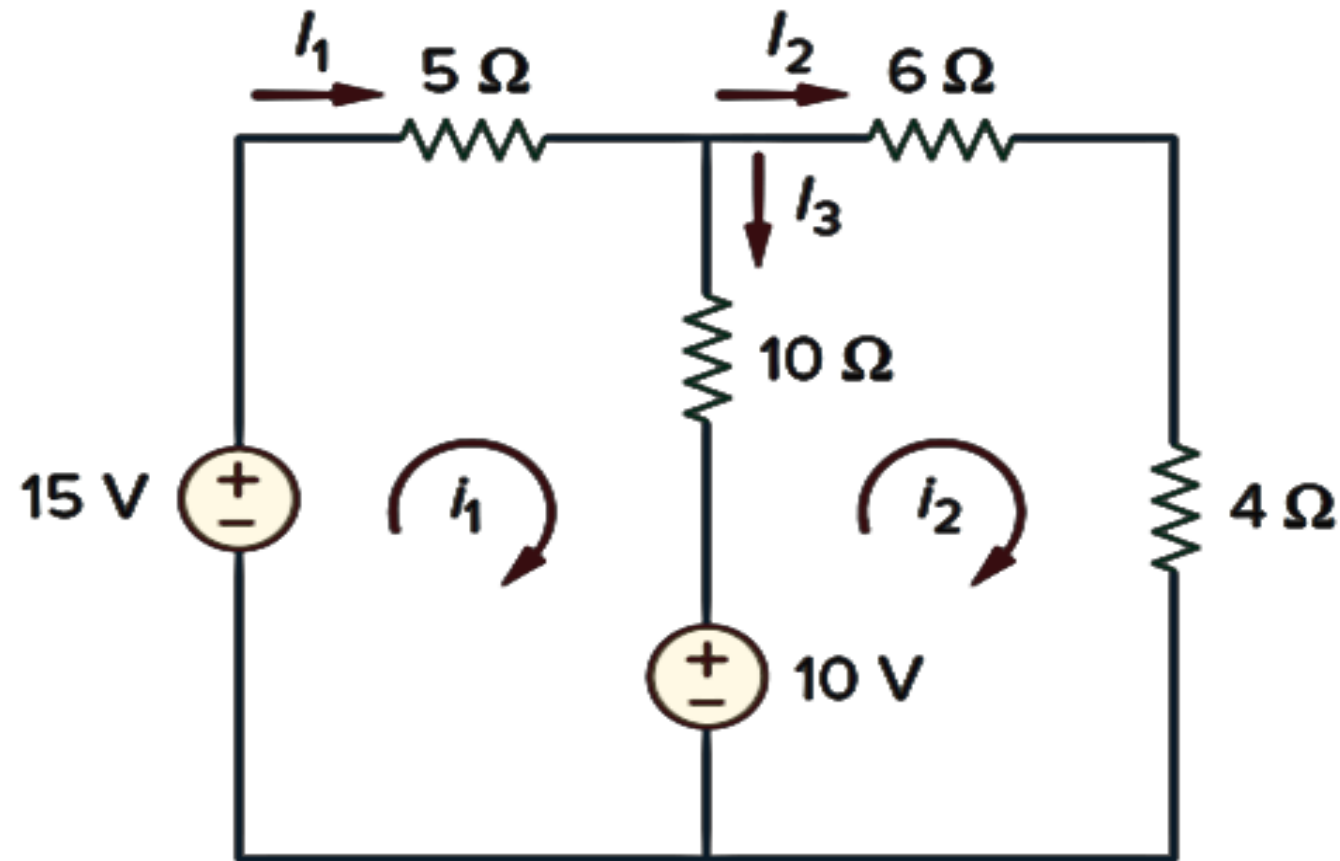




EXERCISE/NUMERICAL ANALYSIS



Q. Find mesh currents i_1 , i_2 , i_3 using mesh analysis.





EXERCISE/NUMERICAL ANALYSIS



Applying KVL in mesh 1.

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

$$\Rightarrow 3i_1 - 2i_2 = 1 \quad (1)$$

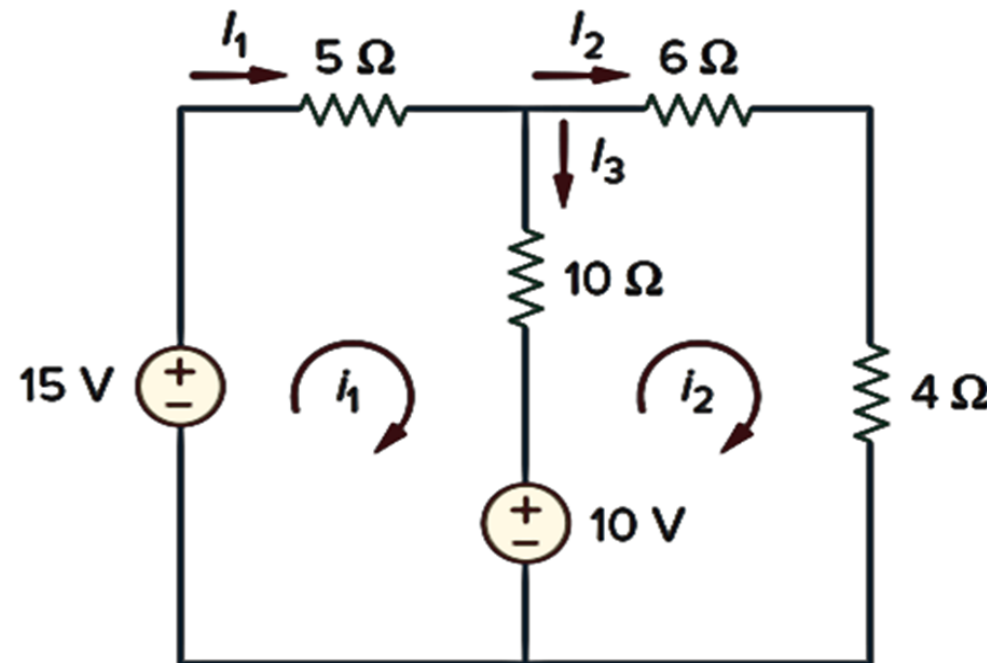
Applying KVL in mesh 2,

$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

$$\Rightarrow i_1 - 2i_2 = -1 \quad (2)$$

Solving above two equations we have

$$i_1 = 1A, i_2 = 1A, i_3 = (i_1 - i_2) = 0A$$





EXERCISE/NUMERICAL ANALYSIS



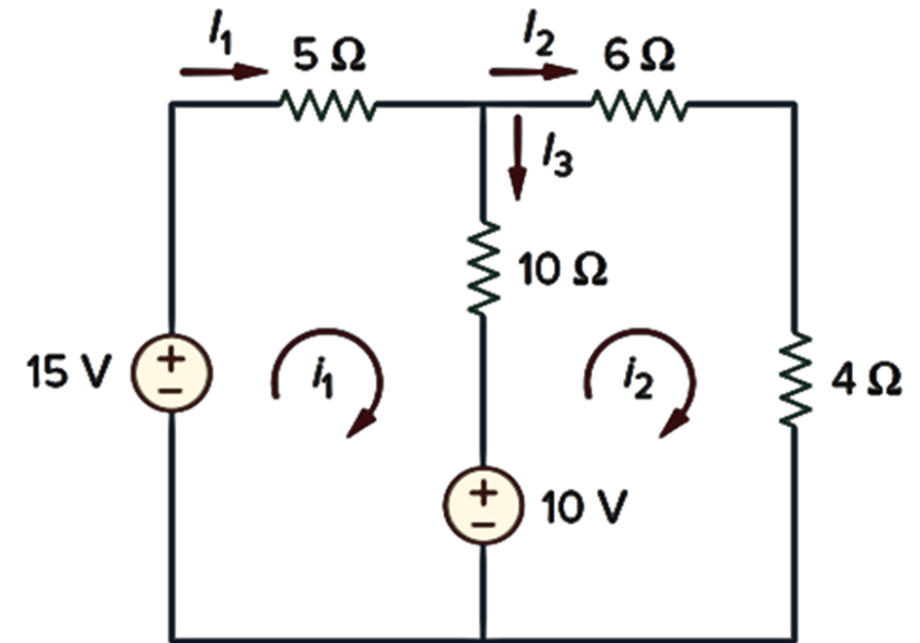
Eqs. (1) and (2) can also be solved using Cramer's rule.

In matrix form we obtain,

$$\begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Calculating determinants,

$$\Delta = \det \begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix} = 6 - 2 = 4$$





EXERCISE/NUMERICAL ANALYSIS

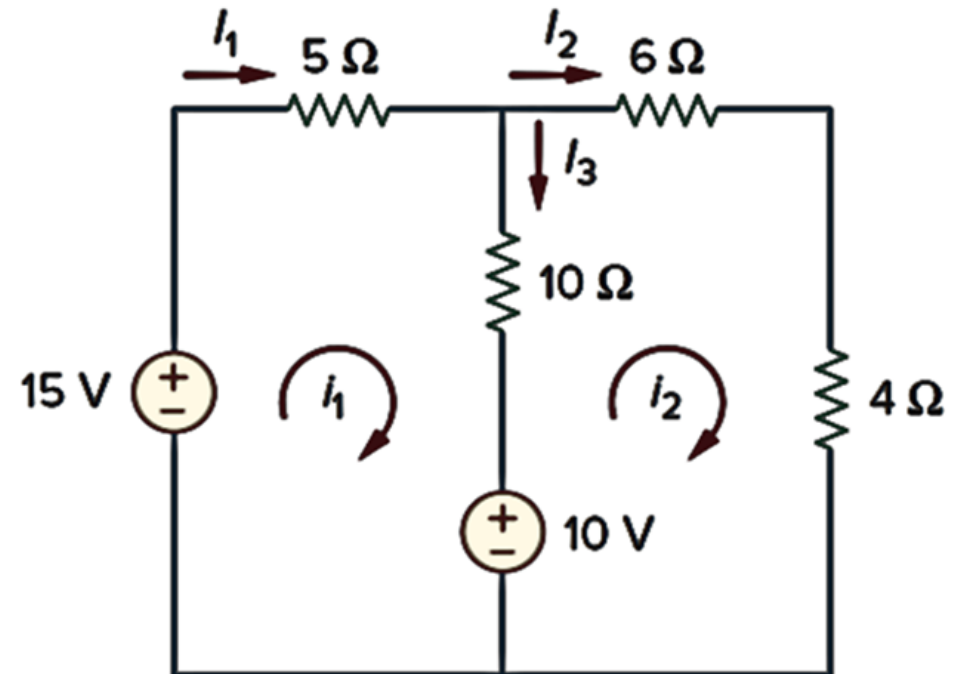


$$\Delta_1 = \det \begin{bmatrix} 1 & -2 \\ 1 & 2 \end{bmatrix} = 2 + 2 = 4$$

$$\Delta_2 = \det \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} = 3 + 1 = 4$$

$$i_1 = \frac{\Delta_1}{\Delta} = 1A, i_2 = \frac{\Delta_2}{\Delta} = 1A$$

$$i_3 = i_1 - i_2 = 0A$$

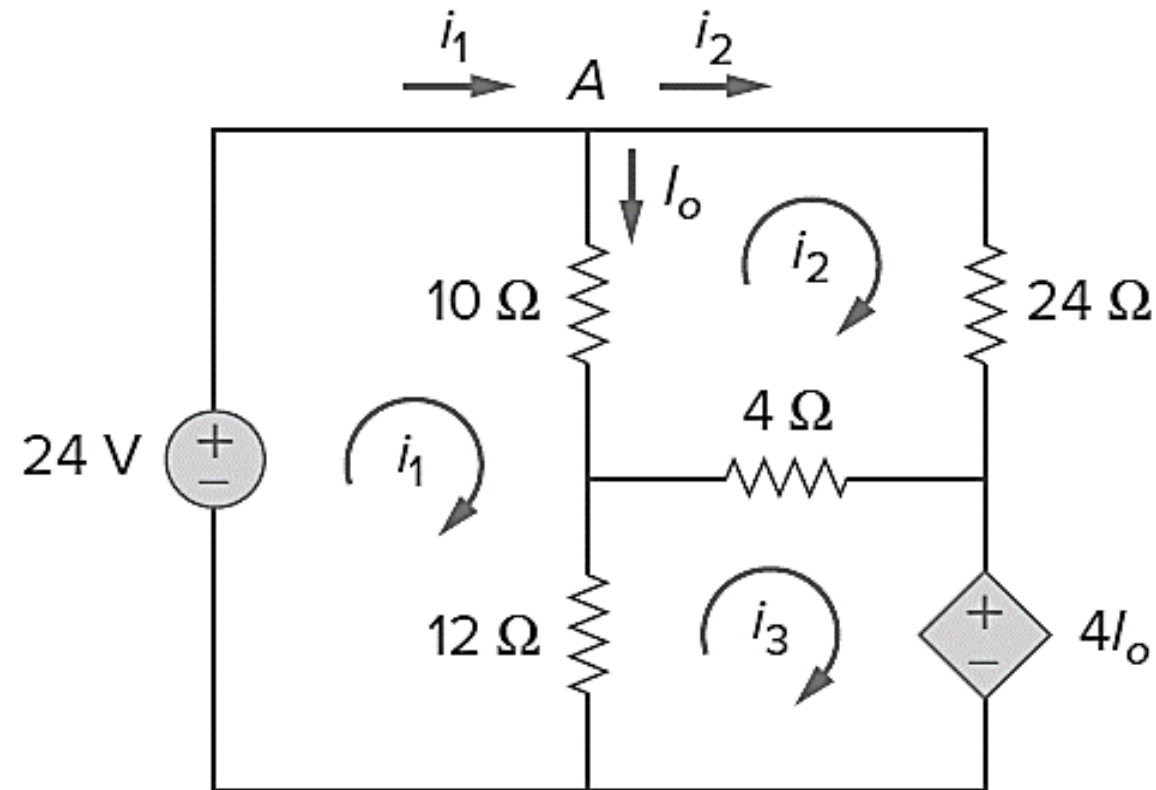




EXERCISE/NUMERICAL ANALYSIS



Q. Find mesh currents i_1 , i_2 , i_3 and i_o using mesh analysis.





EXERCISE/NUMERICAL ANALYSIS



Applying KVL in mesh 1.

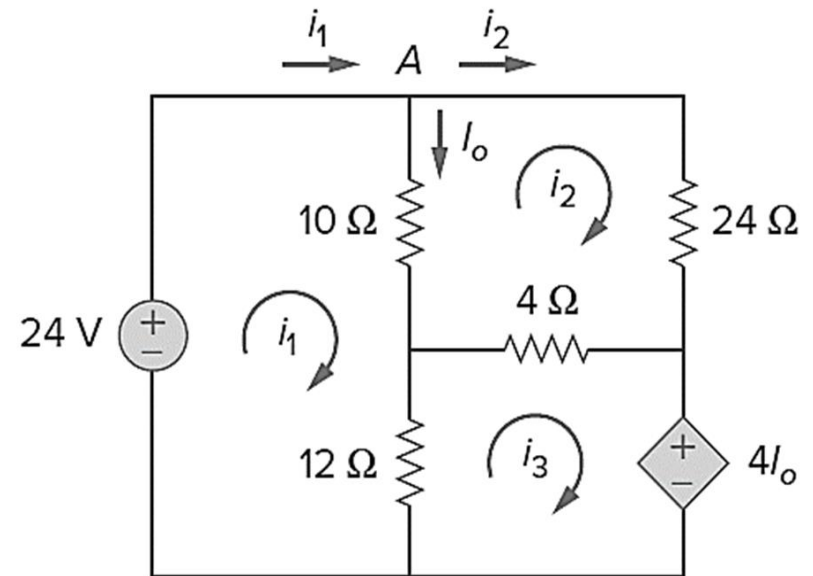
$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0 \Rightarrow 11i_1 - 5i_2 - 6i_3 = 12 \quad (1)$$

Applying KVL in mesh 2,

$$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0 \Rightarrow -5i_1 + 19i_2 - 2i_3 = 0 \quad (2)$$

Applying KVL in mesh 3

$$4I_o + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$





EXERCISE/NUMERICAL ANALYSIS



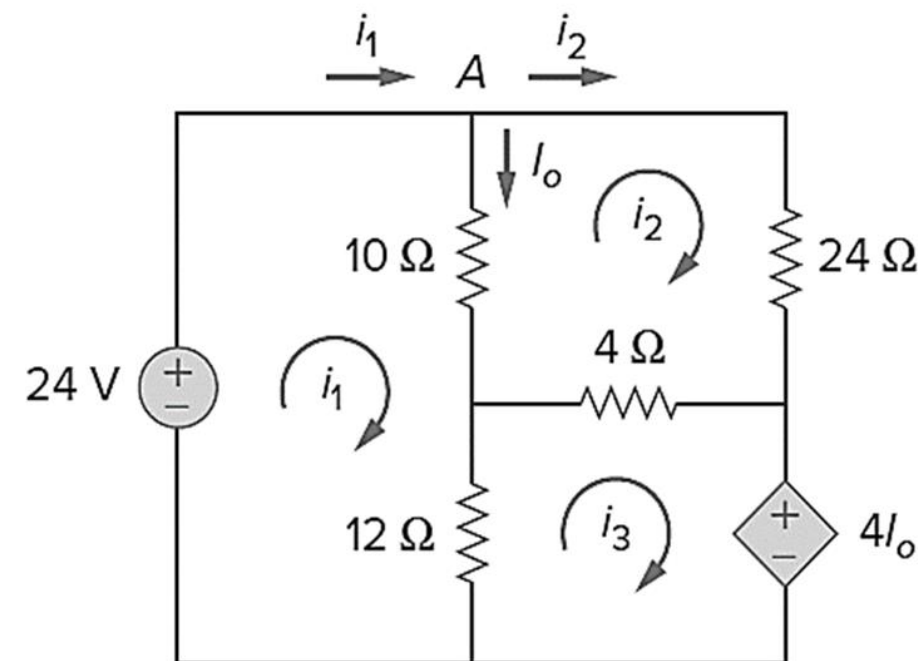
But at node A, $I_o = i_1 - i_2$, so that

$$\Rightarrow 4(i_1 - i_2) + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

$$\Rightarrow -i_1 - i_2 + 2i_3 = 0 \quad (3)$$

Using cramer's' rule to solve 3 unknown variables and writing in matrix form.

$$\begin{bmatrix} 11 & -5 & -6 \\ -5 & 19 & -2 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$$





EXERCISE/NUMERICAL ANALYSIS

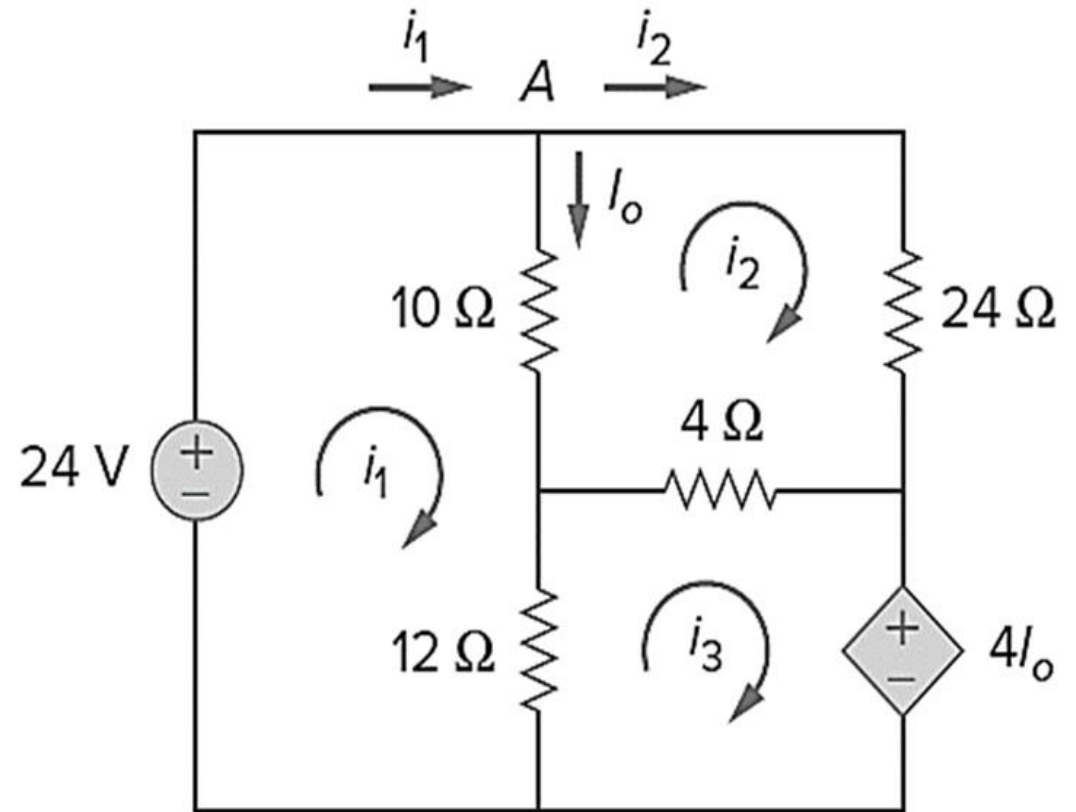


$$\Delta = \det \begin{bmatrix} 11 & -5 & -6 \\ -5 & 19 & -2 \\ -1 & -1 & 2 \end{bmatrix} = 192$$

$$\Delta_1 = \det \begin{bmatrix} 12 & -5 & -6 \\ 0 & 19 & -2 \\ 0 & -1 & 2 \end{bmatrix} = 432$$

$$\Delta_2 = \det \begin{bmatrix} 11 & 12 & -6 \\ -5 & 0 & -2 \\ -1 & 0 & 2 \end{bmatrix} = 144$$

$$\Delta_3 = \det \begin{bmatrix} 11 & -5 & 12 \\ -5 & 19 & 0 \\ -1 & -1 & 0 \end{bmatrix} = 288$$

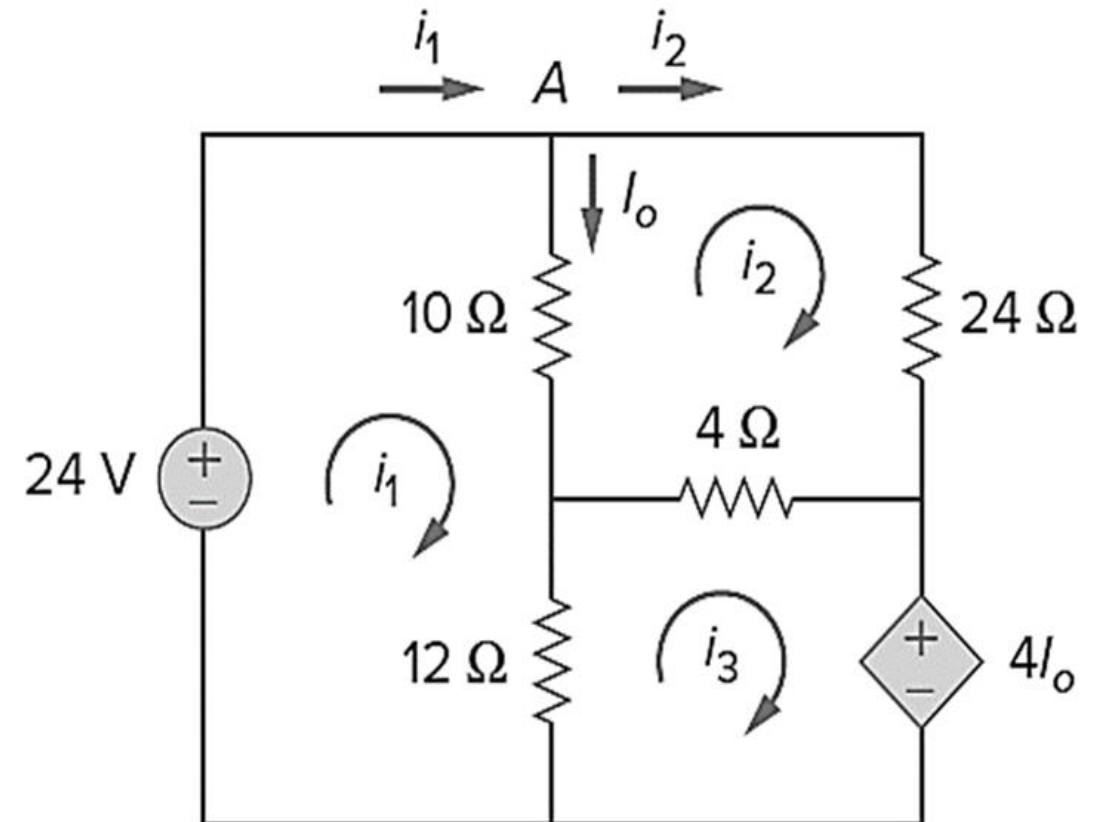




EXERCISE/NUMERICAL ANALYSIS



$$i_1 = \frac{\Delta_1}{\Delta} = 2.25A, i_2 = \frac{\Delta_2}{\Delta} = 0.75A$$
$$i_3 = \frac{\Delta_3}{\Delta} = 1.5A$$

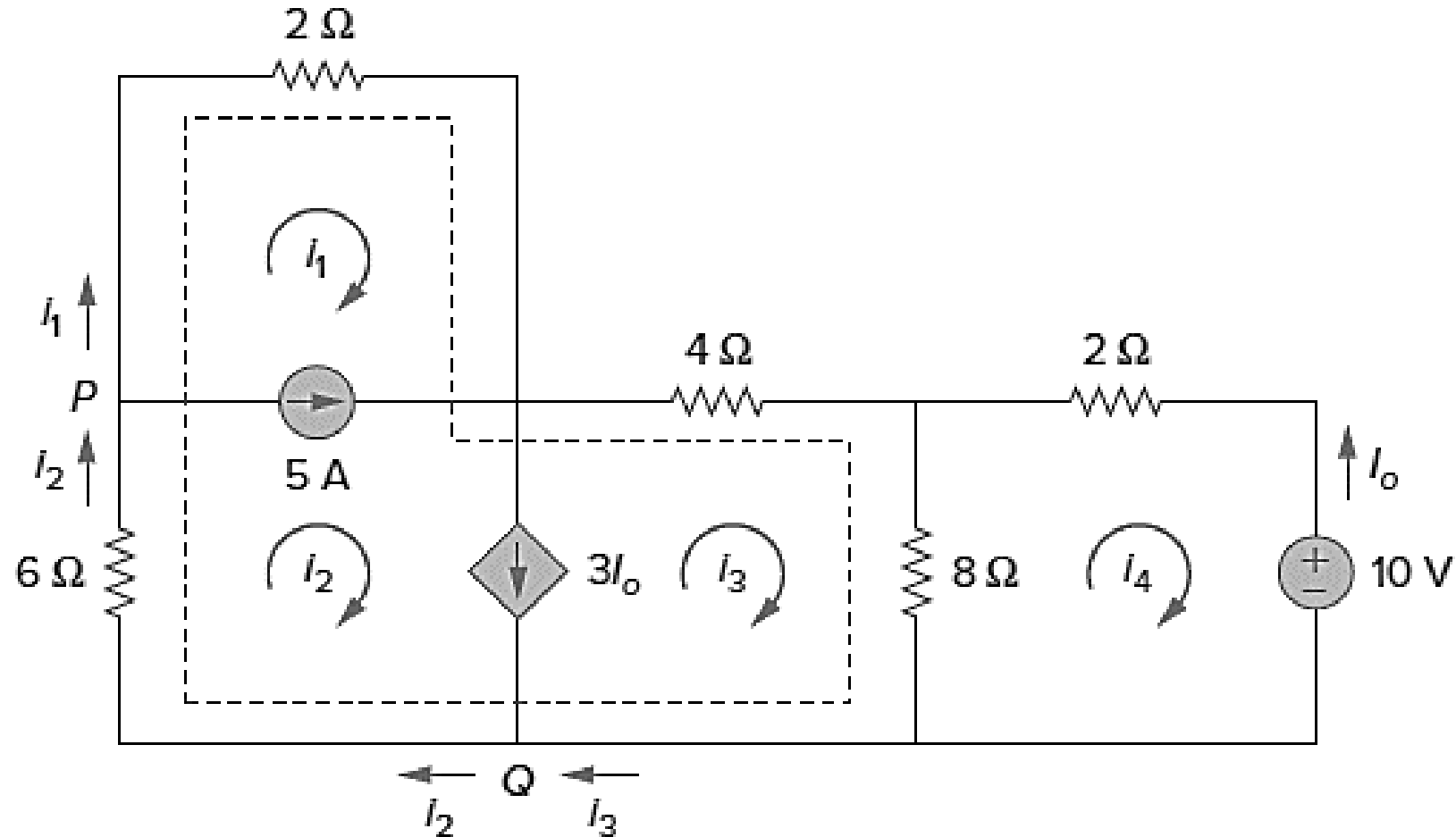




EXERCISE/NUMERICAL ANALYSIS



Q. Find mesh currents i_1 , i_2 , i_3 and i_4 using mesh analysis.





EXERCISE/NUMERICAL ANALYSIS



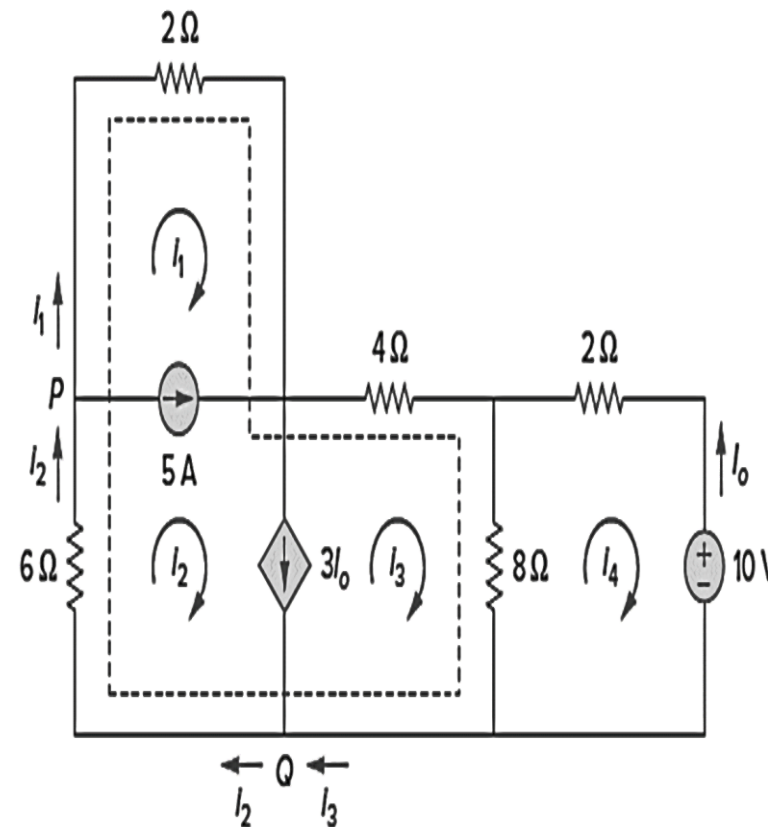
Soln:

Note : meshes 1 and 2 form a supermesh because they have an independent current source in common.

Note : meshes 2 and 3 form another supermesh because they have a dependent current source in common.

The two supermeshes intersect and form a larger supermesh as shown. Applying KVL to the larger supermesh,

$$2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$





EXERCISE/NUMERICAL ANALYSIS



$$\Rightarrow i_1 + 3i_2 + 6i_3 - 4i_4 = 0 \quad \dots(1)$$

Applying KCL to node P: $i_2 = i_1 + 5 \quad \dots(2)$

Applying KCL to node Q: $i_2 = i_3 + 3I_o$

but $I_o = -i_4$

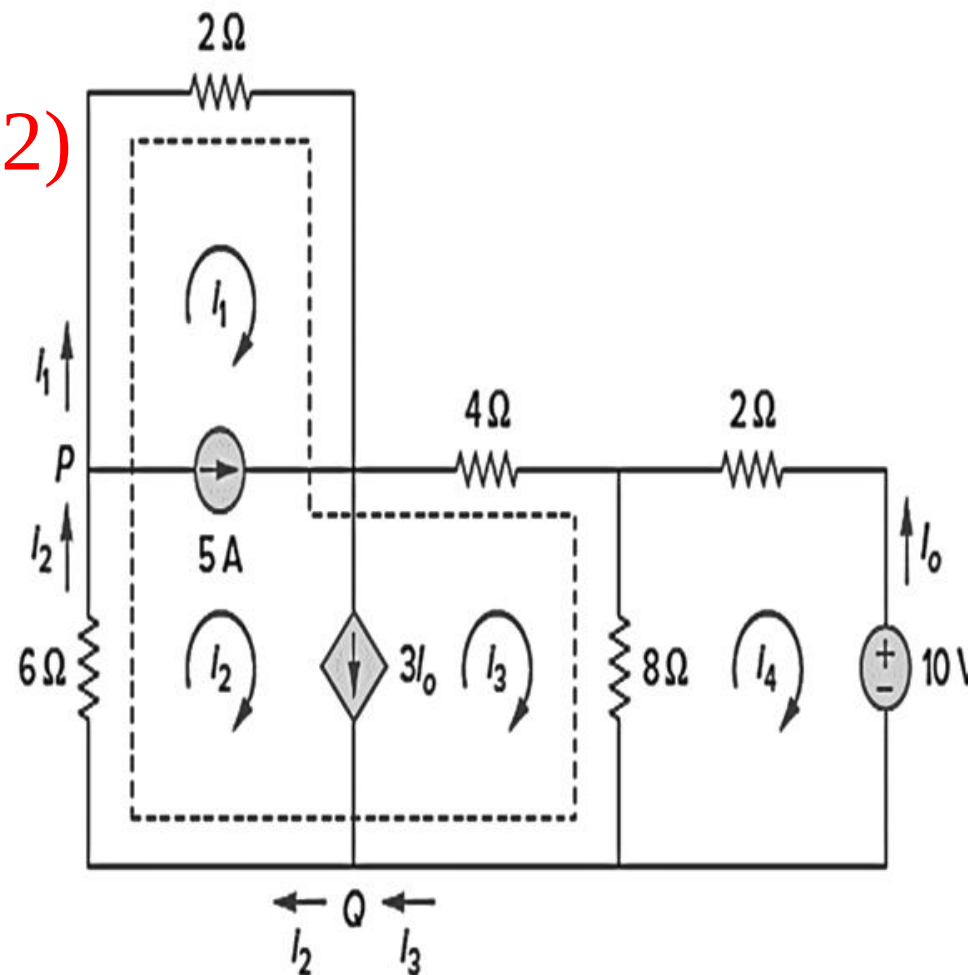
Hence, $i_2 = i_3 - 3i_4 \quad \dots(3)$

Applying KVL in mesh 4,

$$\Rightarrow 2i_4 + 8(i_4 - i_3) + 10 = 0 \quad \dots(4)$$

Solving (1),(2),(3) and (4) we obtain

$$i_1 = -7.5A, i_2 = -2.5A, i_3 = 3.93A \text{ and } i_4 = 2.143A$$

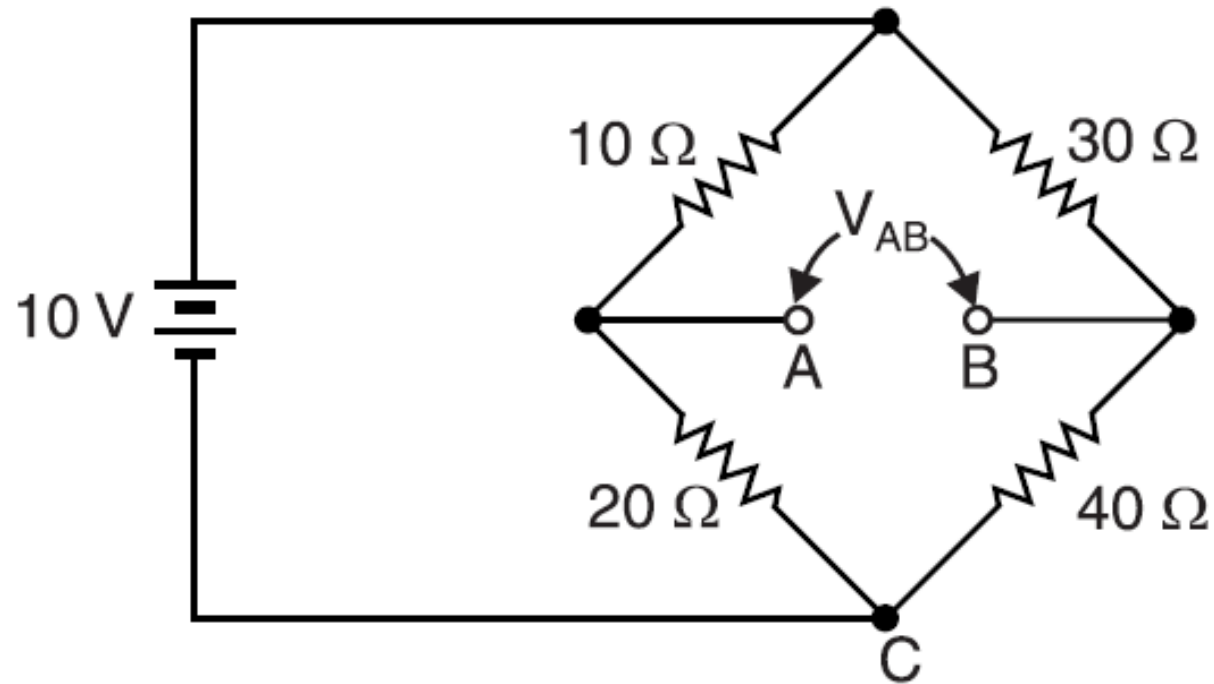




EXERCISE



Q. Find the voltage V_{AB} in the circuit shown below.

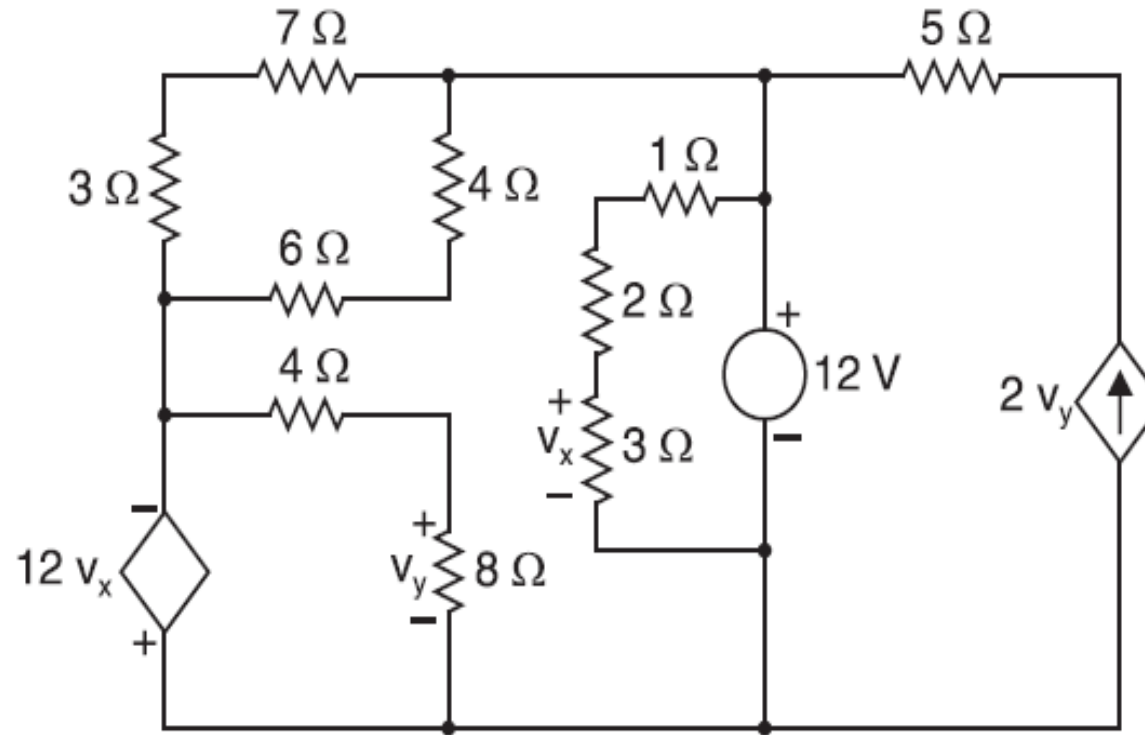




EXERCISE



Q. Find the voltage V_{AB} in the circuit shown below.

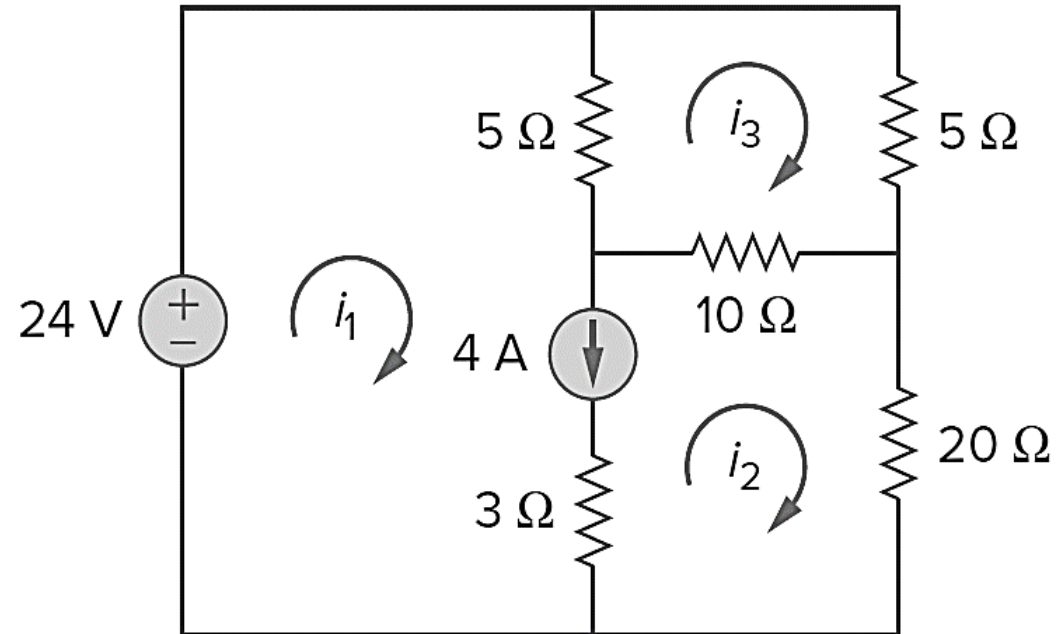




EXERCISE



Q. Use mesh analysis to determine mesh currents.

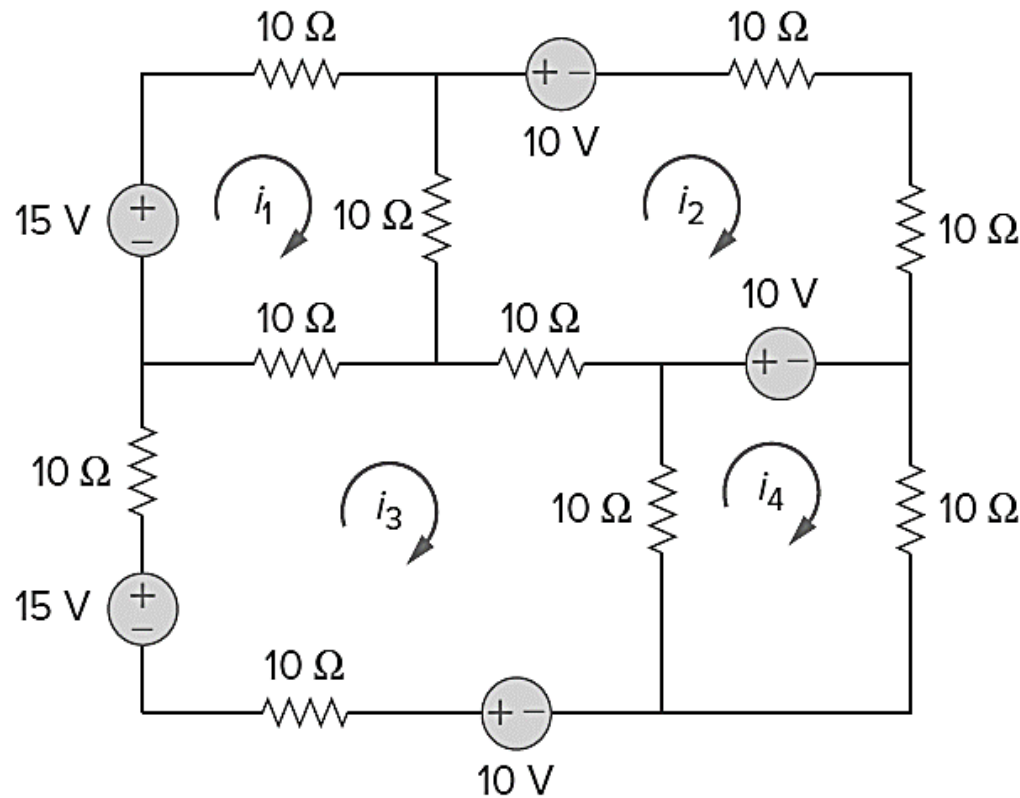




EXERCISE



Q. Write the mesh-current equations for the circuit.

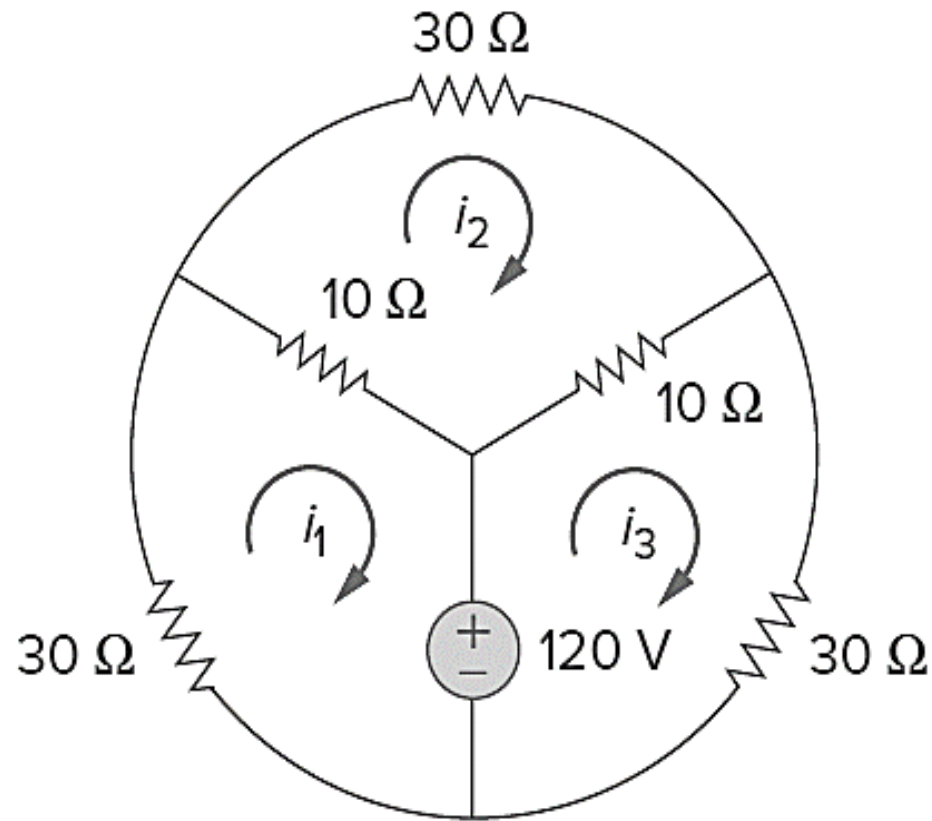




EXERCISE



Q. Write the value of mesh-currents in the circuit.



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- [4] Google Images.(2019). Retrieved from <http://www.googleimages.com>.